Classical and Bayesian estimation of multicomponent stress-strength reliability

Weibull populations under block progressive hybrid censoring and upper k-record sampling

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This report accompanies the fully executed Jupyter notebook. It presents the distribution, principal derivations, unrestricted and ordered inference, 48-scenario simulation, two published-data applications, uncertainty assessments and source-use decisions. The notebook and derivations report contain the complete mathematical development and executable calculation.

The main reliability target is the probability that at least three of five independent component strengths exceed one common random stress. Strength measurements are progressively censored in independent blocks; stress observations are upper k-records stopped at a prescribed record count.

Assigned requirement	Evidence
MLEs and restricted MLEs	Profile optimization and boundary pooling
Joint Bayesian prior	Dependent gamma-Dirichlet prior
Unrestricted and restricted Bayes	Quadrature, posterior sampling and intervals
Complete simulation study	48 scenarios x 300 replications
Suitable-data validation	Jute and carbon-fibre data with calibrated checks

The reported probabilities concern same-unit laboratory comparisons of breaking-strength populations. Neither application supplies measured in-service operating loads. The jute fit limitation is reported explicitly; the additional carbon dataset supports the Weibull validation.

1. Model, notation and assumptions

There are K independent component strengths $X_1, \ldots, X_K$ and a single stress Y shared by the components. The system works if at least s components exceed Y. All strengths are identically distributed and independent of Y. Strength observations used for estimation come from independent blocks drawn from this same population; a block is a testing facility, not an extra component in the system. We use $K = 5$ and $s = 3$ for the main experiment. The record order k is different from the number of components K.

For a positive variable Z, the rate parameterization of the Weibull distribution is

$$f(z; \alpha, \lambda) = \alpha\lambda z^{\alpha - 1} e^{-\lambda z^{\alpha}}$$

$$F(z) = 1 - e^{-\lambda z^{\alpha}}$$

Thus

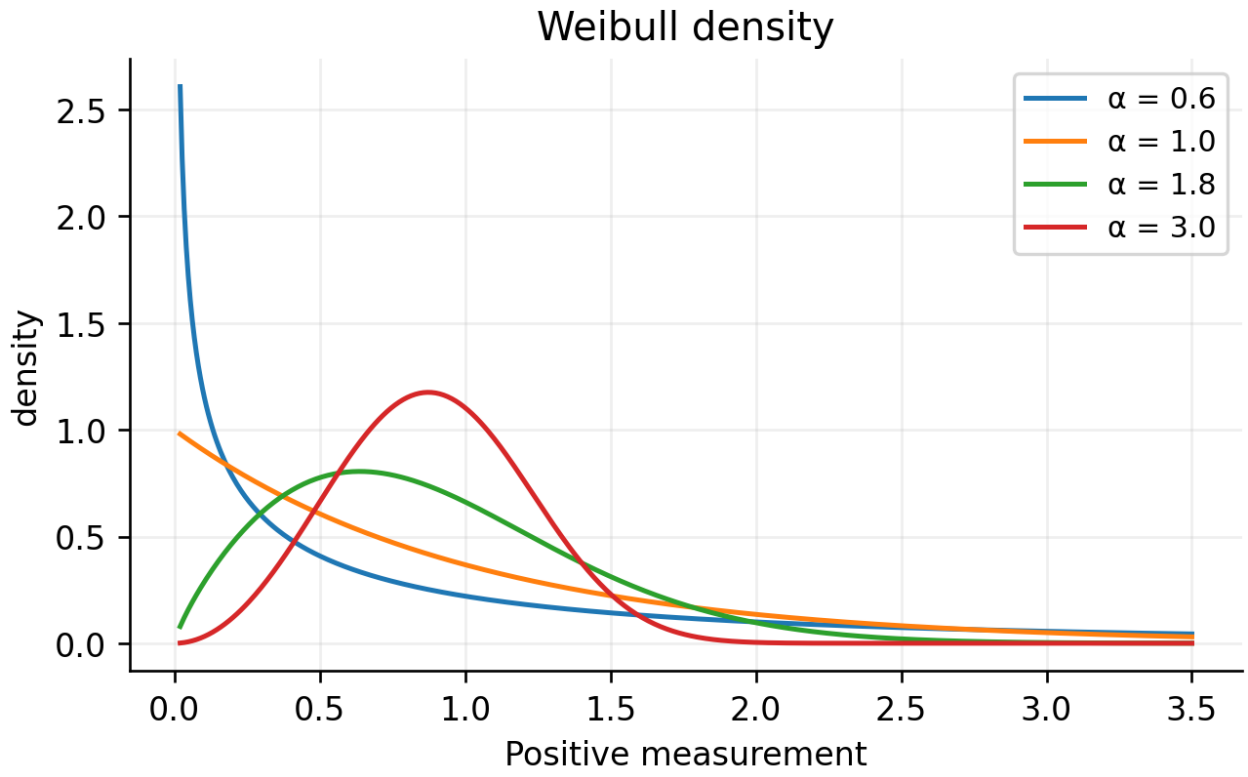
$$S(z) = e^{-\lambda z^{\alpha}}, \qquad h(z) = \alpha\lambda z^{\alpha - 1}$$

$$F^{-1}(u) = \left[\frac{-\log(1-u)}{\lambda}\right]^{1/\alpha}$$

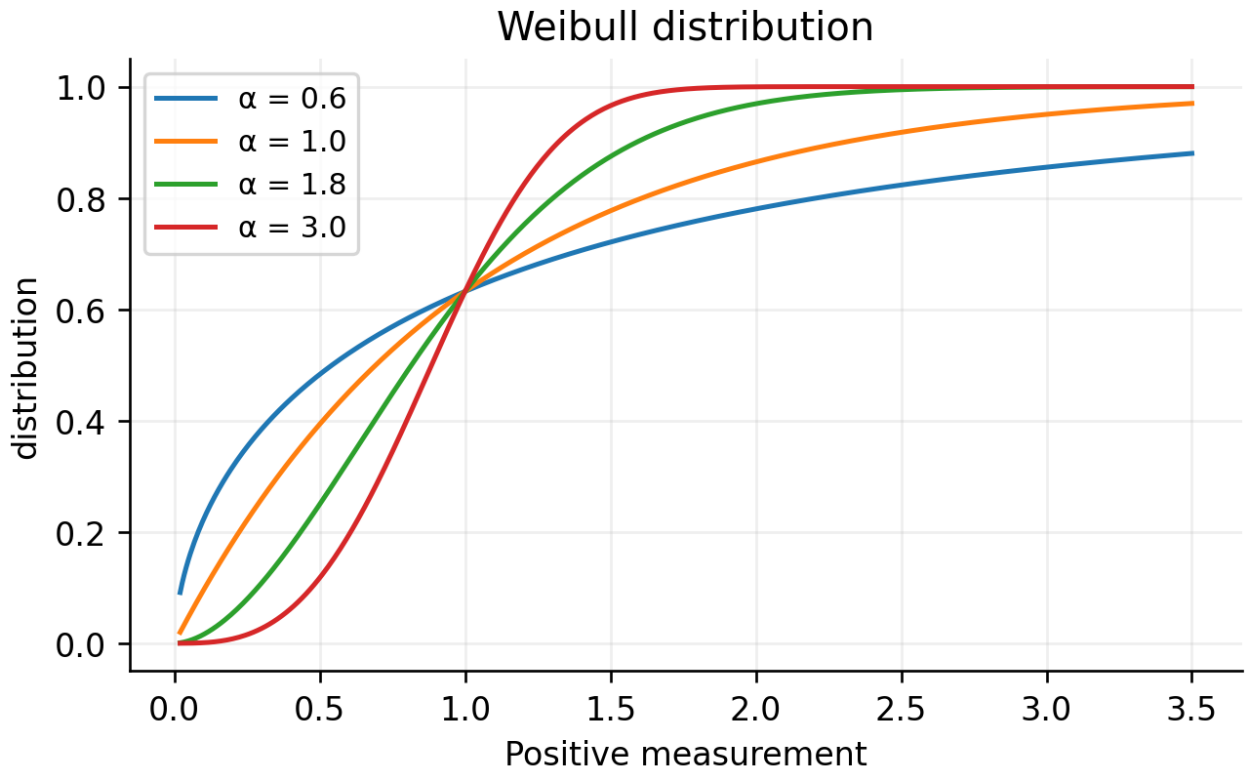
The conventional Weibull scale is $\eta = \lambda^{-1/\alpha}$. Calling λ a rate avoids reversing the stochastic ordering. We assume $X \sim \text{Weibull}(\alpha, a)$ and $Y \sim \text{Weibull}(\alpha, b)$, with $\alpha, a, b > 0$. Their common shape yields a tractable reliability formula and is checked on the complete application data. Under the increasing transformation $z \mapsto z^{\alpha}$, strengths become exponential with rate a and stress becomes exponential with rate b.

The natural engineering restriction is $a \leq b$: strength has larger survival at every positive threshold. Equivalently, $\eta_X \geq \eta_Y$. This restriction is an assumption about populations, not a conclusion automatically established by the data. Simulation deliberately includes $a > b$ to quantify the effect of imposing a false restriction.

Weibull density, distribution, survival and hazard

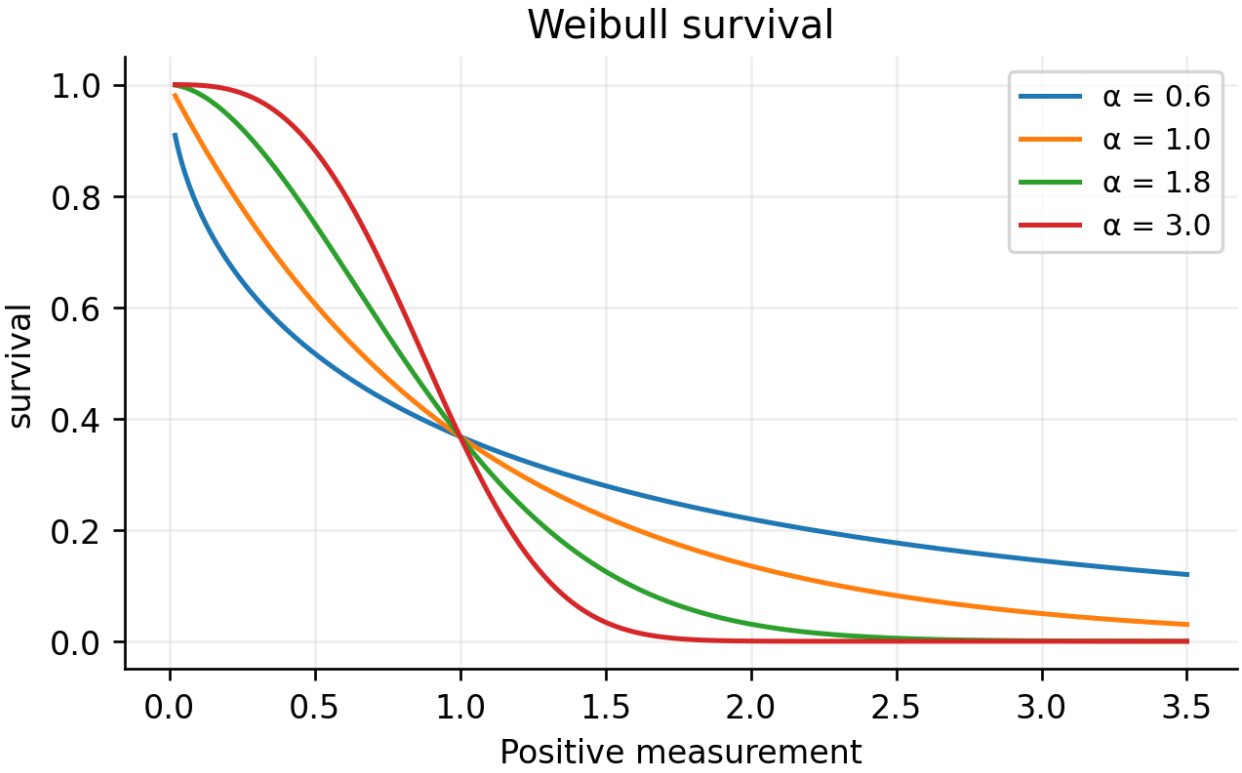


Weibull density

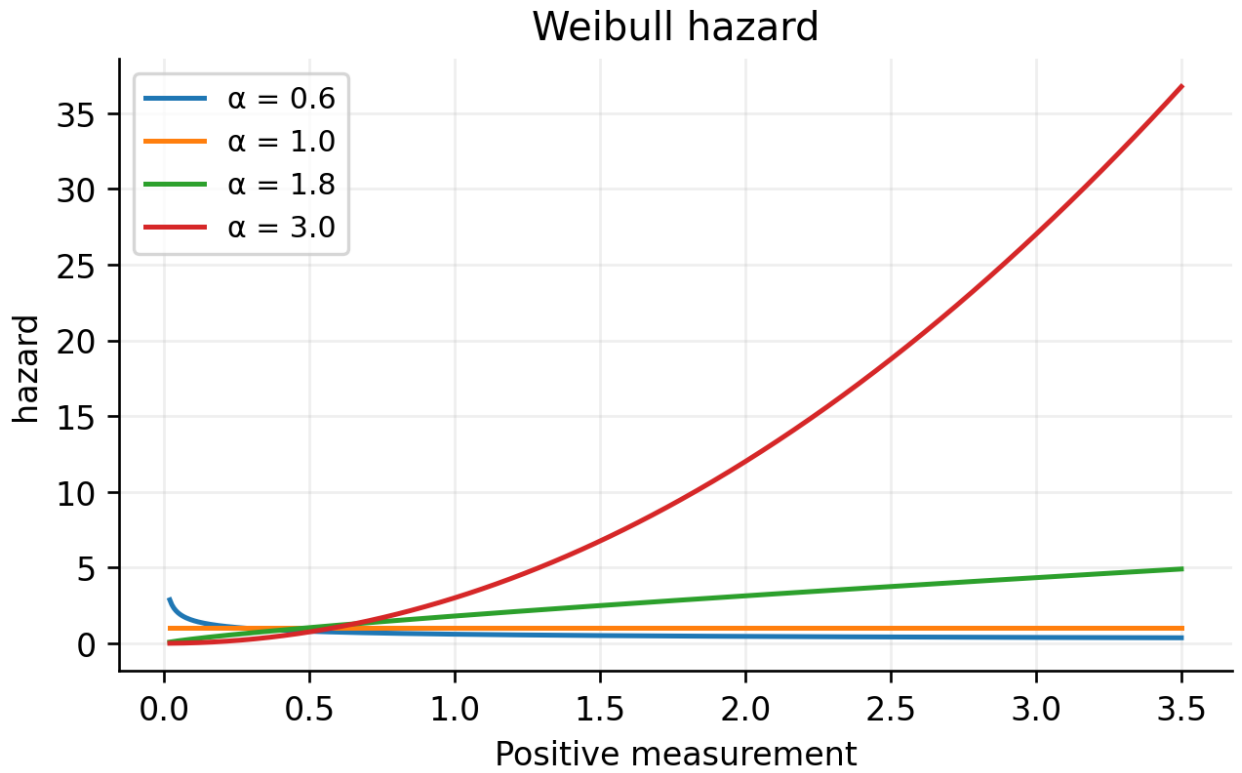


Weibull distribution

Weibull density, distribution, survival and hazard



Weibull survival



Weibull hazard

2. Reliability derivation

Conditional on $Y = y$, the number of components surviving is binomial with success probability $S_X(y)$. The conditional binomial probability must be integrated over the common stress:

$$R_{s,K} = \int_0^\infty \sum_{j=s}^{K} \binom{K}{j} S_X(y)^j [1 - S_X(y)]^{K-j} f_Y(y)\, dy$$

Set $\rho = b/a$ and $u = \exp(-ay^\alpha)$. The transformed stress density satisfies

$$f_Y(y)\, dy = -\rho u^{\rho - 1}\, du$$

$$R_{s,K} = \rho \sum_{j=s}^{K} \binom{K}{j} \int_0^1 u^{\rho + j - 1} (1 - u)^{K-j}\, du$$

$$R_{s,K} = \rho \sum_{j=s}^{K} \binom{K}{j} B(\rho + j, K - j + 1)$$

An equivalent and more stable expression follows from exponential order statistics. At least s strengths exceed stress exactly when Y^α is below the $(K - s + 1)$-th smallest transformed strength. That order statistic is the sum of independent exponentials with rates $Ka, (K-1)a, \ldots, sa$. Its Laplace transform at b is the product of $ja/(ja + b)$. Therefore

$$R_{s,K} = 1 - \prod_{j=s}^{K} \frac{j}{j + \rho}$$

Useful limiting checks are

$$R_{1,1} = \frac{b}{a + b}, \qquad R_{K,K} = \frac{b}{Ka + b}$$

$$\lim_{\rho \to 0} R_{s,K} = 0, \qquad \lim_{\rho \to \infty} R_{s,K} = 1$$

Reliability decreases as s increases. For fixed a and b, reliability is independent of the common shape α. The shape still affects rate estimates through the likelihood. For $s = 3$, $K = 5$ and $a = b$,

$$R_{3,5} = 1 - \frac{3}{4}\frac{4}{5}\frac{5}{6} = \frac{1}{2}$$

The common stress makes component success indicators dependent after Y is integrated out:

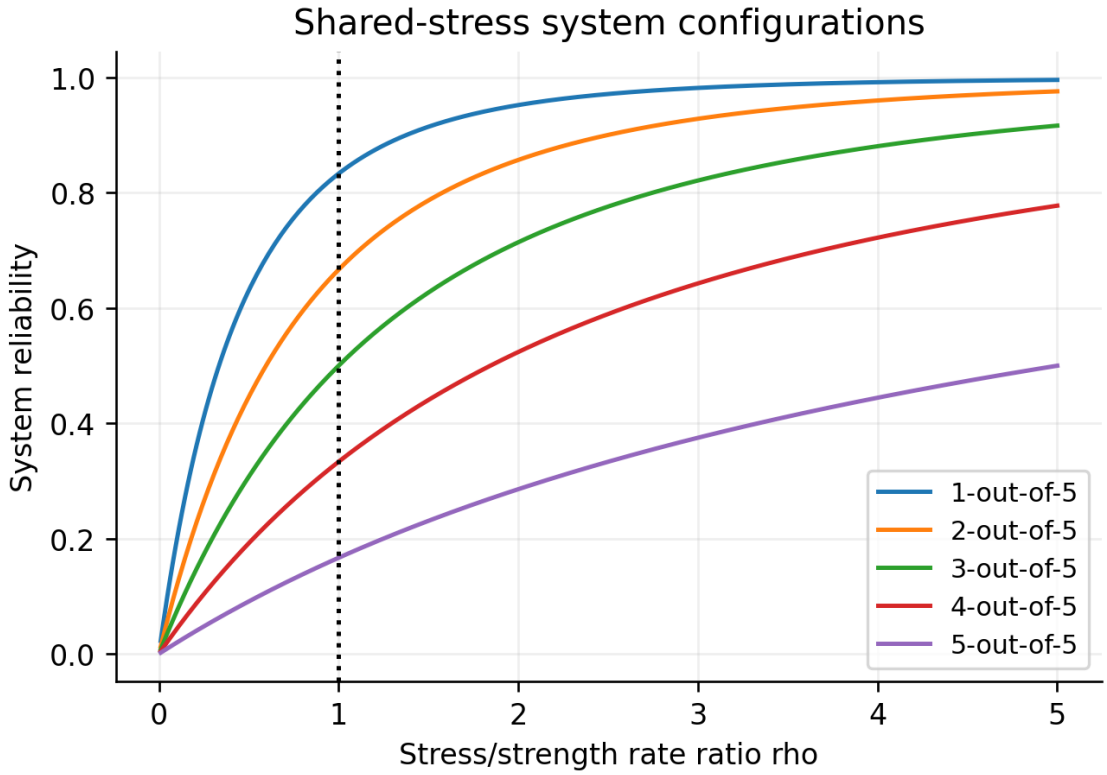
$$\mathrm{Cov}(I[X_i > Y], I[X_j > Y]) = \frac{b}{b+2a} - \left(\frac{b}{b+a}\right)^2 > 0$$

Consequently replacing $S_X(y)$ by $P(X > Y)$ in the conditional binomial sum computes a different model. This project uses the shared-stress integral rather than the unconditional binomial expression in equations (2.5)-(2.7) of Newer (2025).

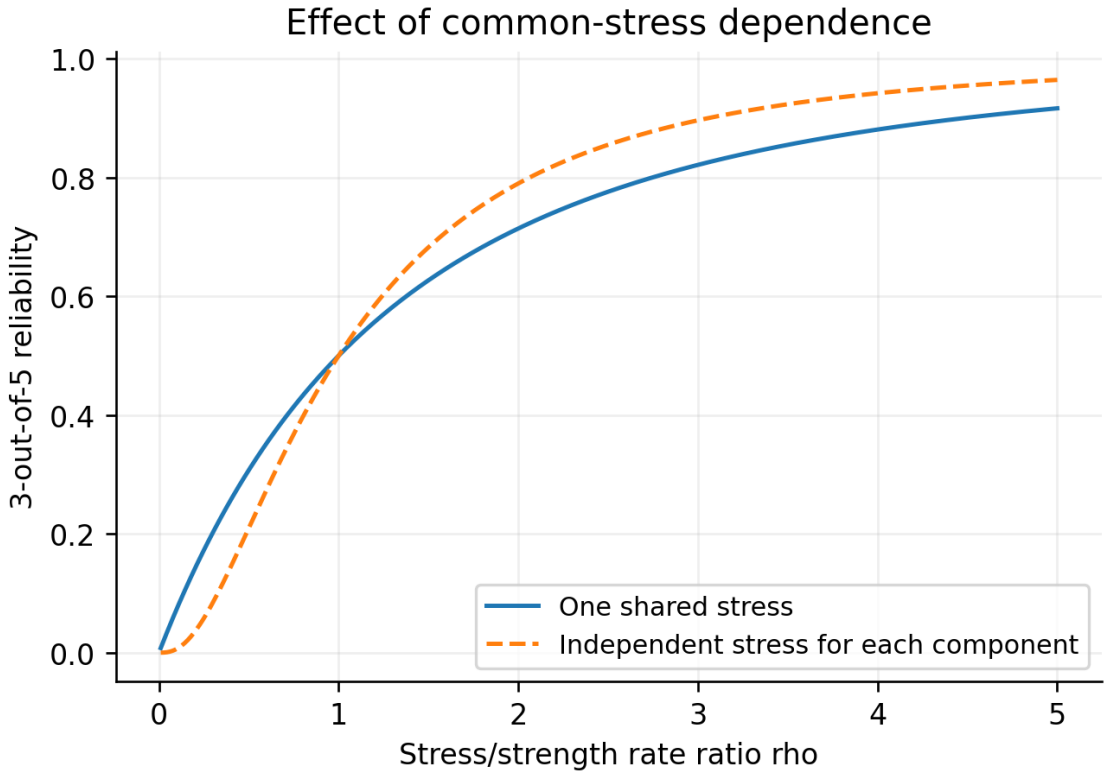
For stable numerical evaluation,

$$R = -\mathrm{expm1}\left(-\sum_{j=s}^{K} \log(1+\rho/j)\right)$$

Shared-stress system reliability



Shared-stress system configurations



Effect of common-stress dependence

3. Strength sampling: block progressive hybrid censoring

In block i, n_i specimens start together. A target m_i, threshold T_i and planned withdrawal vector $(R_{i1}, \ldots, R_{im_i})$ are specified before testing, with $\sum_j R_{ij} = n_i - m_i$. At observed failure x_{ij}, the prescribed number of survivors is selected uniformly for withdrawal. Withdrawals are noninformative and block assignment is random.

The principal Type-I scheme stops at $\tau_i = \min(x_{im_i}, T_i)$. If the target is reached before T_i, all remaining survivors are withdrawn at the last failure. Otherwise $d_i < m_i$ failures are observed. Using the realized withdrawal counts r_{ij}, the number of terminal survivors is

$$q_i = n_i - d_i - \sum_{j=1}^{d_i} r_{ij}$$

The block likelihood, up to design factors independent of the model parameters, is

$$L_i \propto \prod_{j=1}^{d_i} f_X(x_{ij}) S_X(x_{ij})^{r_{ij}} S_X(\tau_i)^{q_i}$$

Here $q_i = 0$ when the target is reached because final survivors are already included in $r_{i,\,d_i}$. There is no double-counting of final withdrawals. A block with zero failures still contributes its terminal survival factor.

The adaptive Type-II alternative matches the main reference's design. It guarantees m_i failures: intermediate withdrawals at failures after T_i are canceled, and all survivors are withdrawn at the final target failure. Its duration may exceed T_i. Its likelihood uses the same expression with $d_i = m_i$, $q_i = 0$ and the actual adapted counts r_{ij}. The notebook separately implements both protocols and records specimen counts and stopping outcomes.

Multiplying independent block likelihoods gives

$$L_X \propto \alpha^d a^d \exp\left[(\alpha - 1)\sum_{i,j} \log x_{ij} - aA(\alpha)\right]$$

$$d = \sum_i d_i, \qquad A(\alpha) = \sum_{i,j} (1 + r_{ij}) x_{ij}^{\alpha} + \sum_i q_i \tau_i^{\alpha}$$

Scheduled removals after termination contribute nothing. The specimen-conservation check is $n_i = d_i + \sum_j r_{ij} + q_i$. Exposure is the sum of transformed failure times and transformed right-censoring times for every starting unit.

4. Stress sampling: upper k-record values

Starting with the first k independent stress observations, record the k-th largest, which is their minimum. Subsequently a record transition occurs whenever a new observation exceeds the current k-th largest; after updating the largest k observations, retain their new minimum. Collect exactly r successive records $y_1 < \cdots < y_r$ and stop at the final prescribed record. Original table positions in the application are an artificial acquisition sequence, not historical timestamps.

On the cumulative-hazard scale $H_Y(y) = -\log S_Y(y)$, the successive k-record increments are independent exponential variables with rate k. One justification is that the k surviving exceedances above the current threshold have independent exponential residual hazards; their next minimum has rate k. The joint density on the hazard scale is $k^r \exp[-kH_Y(y_r)]$ over the ordered region. Transforming back using $H'_Y(y) = h_Y(y)$ gives

$$L_Y = k^r \exp[-kH_Y(y_r)] \prod_{j=1}^{r} h_Y(y_j) = k^r S_Y(y_r)^k \prod_{j=1}^{r} \frac{f_Y(y_j)}{S_Y(y_j)}$$

For Weibull observations,

$$L_Y \propto \alpha^r b^r \exp\left[(\alpha - 1)\sum_{j=1}^{r} \log y_j - bB(\alpha)\right]$$

$$B(\alpha) = k y_r^{\alpha}$$

The records are dependent and are never treated as an independent Weibull sample. Simulation uses

$$y_j = \left(\frac{\sum_{l=1}^{j} E_l}{kb}\right)^{1/\alpha}, \qquad E_l \sim \text{Exp}(1)$$

Record waiting times are not included in this marginal likelihood. The notebook retains acquisition positions for audit only. Including waiting times requires a different joint likelihood; the inter-record expression in the main reference is not copied into this marginal analysis.

5. Joint likelihood and maximum likelihood estimates

Define $q = d + r$ and

$$C = \sum_{i,j} \log x_{ij} + \sum_{j=1}^{r} \log y_j$$

The parameter-dependent log-likelihood is

$$\ell(\alpha, a, b) = q \log \alpha + d \log a + r \log b + (\alpha - 1)C - aA(\alpha) - bB(\alpha)$$

The score equations are

$$\ell_a = d/a - A(\alpha), \qquad \ell_b = r/b - B(\alpha)$$

$$\ell_\alpha = q/\alpha + C - aA'(\alpha) - bB'(\alpha)$$

For fixed α, the unique positive rate estimates are

$$\hat{a}(\alpha) = \frac{d}{A(\alpha)}, \qquad \hat{b}(\alpha) = \frac{r}{B(\alpha)}$$

The unrestricted shape estimate maximizes

$$\ell_p(\alpha) = q \log \alpha - d \log A(\alpha) - r \log B(\alpha) + (\alpha - 1)C + \text{constant}$$

The profile score is

$$\ell_p'(\alpha) = \frac{q}{\alpha} + C - d\frac{A'(\alpha)}{A(\alpha)} - r\frac{B'(\alpha)}{B(\alpha)}$$

Write $A(\alpha) = \sum_l w_l \exp(\alpha \log z_l)$. The second derivative of $\log A$ is the weighted variance of $\log z_l$, with weights proportional to $w_l z_l^\alpha$. Since $\log B$ is linear in α,

$$\ell_p''(\alpha) = -q/\alpha^2 - d\,\mathrm{Var}_{w_l z_l^\alpha / A}(\log z_l) < 0$$

Distinct records ensure a finite interior maximum: the score diverges positively near zero and is negative eventually because the largest record is larger than earlier records. This proof assumes $d > 0$ and $r \geq 2$. If $d = 0$, the unrestricted strength-rate supremum can occur at $a = 0$; the software reports this instead of presenting a finite interior MLE. All delivered experiments meet the stated conditions.

For fixed α under $a \le b$, either the unrestricted pair satisfies the order or the optimum lies on the boundary. On the boundary $a = b = \lambda$, differentiation yields $\lambda = q/(A + B)$. Therefore

$$(\tilde{a}, \tilde{b}) = (d/A,\ r/B) \quad \text{when} \quad d/A \le r/B$$

$$\tilde{a} = \tilde{b} = q/(A + B) \quad \text{when} \quad d/A > r/B$$

Reoptimize α with this pair at every trial shape. Simply sorting the unrestricted estimates, or pooling at the unrestricted shape without reoptimization, is not the restricted MLE. On the pooled branch, the second derivative is

$$\ell''_{p,O}(\alpha) = -\frac{q}{\alpha^2} - q\frac{d^2}{d\alpha^2}\log[A(\alpha) + B(\alpha)] < 0$$

The score agrees across a pooling transition. This gives a unique profile maximum under the same conditions. Reliability estimates follow by invariance:

$$\widehat{R} = R(\hat{a}, \hat{b}), \qquad \widetilde{R} = R(\tilde{a}, \tilde{b})$$

6. Observed information and intervals

Differentiate exposures directly:

$$A' = \sum_l w_l z_l^\alpha \log z_l$$

$$A'' = \sum_l w_l z_l^\alpha (\log z_l)^2$$

$$B' = B \log y_r, \qquad B'' = B(\log y_r)^2$$

The negative Hessian in the parameter order (α, a, b) is

Observed-information matrix in the parameter order (α, a, b):

$q/\alpha^2 + aA'' + bB''$	A'	B'
A'	d/a^2	0
B'	0	r/b^2

At an unrestricted MLE, the covariance approximation is

$$V = I(\widehat{\alpha}, \widehat{a}, \widehat{b})^{-1}$$

For parameter θ_j, a positive 95% confidence interval is

$$\theta_{j,\mathrm{L}} = \exp\left[\log \widehat{\theta}_j - 1.96 \frac{\sqrt{V_{jj}}}{\widehat{\theta}_j}\right]$$

$$\theta_{j,\mathrm{U}} = \exp\left[\log \widehat{\theta}_j + 1.96 \frac{\sqrt{V_{jj}}}{\widehat{\theta}_j}\right]$$

For the log-rate ratio, define

$$\psi = \log \rho = \log b - \log a, \qquad g = \left(0, -\tfrac{1}{a}, \tfrac{1}{b}\right)^{\mathsf{T}}$$

Its estimated standard error and ratio endpoints are

$$\mathrm{SE}(\widehat{\psi}) = \sqrt{g^{\mathsf{T}} V g}$$

$$\rho_{\mathrm{L}} = \exp[\widehat{\psi} - 1.96\mathrm{SE}(\widehat{\psi})]$$

$$\rho_{\mathrm{U}} = \exp[\widehat{\psi} + 1.96\mathrm{SE}(\widehat{\psi})]$$

Transform through the increasing reliability function:

$$[R(1, \rho_{\mathrm{L}}),\ R(1, \rho_{\mathrm{U}})]$$

This construction keeps parameter endpoints positive and reliability endpoints between zero and one.

A normal approximation at a pooled restricted MLE is nonregular. The application instead reports percentile parametric bootstrap intervals, regenerating complete strength blocks, withdrawals, hybrid stopping and stress k-records under the fitted model and refitting each replicate. Bootstrap intervals at an order boundary can have point mass at $R = 1/2$; they are descriptive finite-sample approximations, not a claim of exact boundary coverage. Their simulation coverage is assessed separately. Bayesian intervals are posterior credible intervals and have a different interpretation from confidence intervals.

7. Dependent joint prior and its derivation

Use $t = a + b$ and $p = a/(a + b)$. Independently assign

$$t \sim \text{Gamma}(a_0, b_0), \quad p \sim \text{Beta}(c_1, c_2), \quad \alpha \sim \text{Gamma}(a_1, b_1)$$

where Gamma is shape-rate. The absolute Jacobian from (t, p) to (a, b) is t. The rate-pair density is

$$C_0 = \frac{b_0^{a_0}\Gamma(c_1 + c_2)}{\Gamma(a_0)\Gamma(c_1)\Gamma(c_2)}$$

$$\pi(a, b) = C_0 e^{-b_0(a+b)}(a + b)^{a_0 - c_1 - c_2} a^{c_1 - 1} b^{c_2 - 1}$$

This is the two-dimensional gamma-Dirichlet family used in Sultana et al. It gives independent Gamma rates only if $a_0 = c_1 + c_2$. The primary choice is

$$(a_0, b_0, c_1, c_2, a_1, b_1) = (2, 0.5, 2, 2, 2, 1)$$

Since $a_0 \neq c_1 + c_2$, dependence is genuine. For $c = c_1 + c_2$,

$$E(a) = \frac{a_0 c_1}{b_0 c}, \quad E(b) = \frac{a_0 c_2}{b_0 c}, \quad \text{Cov}(a, b) = \frac{a_0 c_1 c_2 (c - a_0)}{b_0^2 c^2 (c + 1)}$$

Here prior means of both rates are 2, prior mean shape is 2, and covariance is 0.8. These are fixed choices rather than hyperparameters fitted to each simulation's true parameters. They are proper, moderately broad priors, not literally noninformative. Prior sensitivity is reported and the application does not recycle the same data into a prior presented as external information.

For the order-restricted model condition this prior on $p \leq 1/2$:

$$\pi_O(\alpha, a, b) = \pi(\alpha, a, b) \frac{I(a \leq b)}{P(p \leq 1/2)}, \qquad P(p \leq 1/2) = I_{1/2}(c_1, c_2)$$

With symmetric $c_1 = c_2 = 2$ the normalizing probability is 1/2. This is a truncated dependent joint prior; sorting draws from a posterior with labelled strength and stress roles would target a different posterior.

8. Posterior, Bayesian estimators and computation

In the original coordinates the posterior kernel is

$$\pi(\alpha, a, b \mid D) \propto \alpha^{q + a_1 - 1} e^{(\alpha - 1)C - b_1 \alpha}$$

$$\times a^{d + c_1 - 1} b^{r + c_2 - 1} (a + b)^{a_0 - c_1 - c_2}$$

$$\times e^{-a(A + b_0) - b(B + b_0)}$$

In (α, t, p), including the Jacobian,

$$\pi(\alpha, t, p \mid D) \propto \alpha^{q + a_1 - 1} e^{(\alpha - 1)C - b_1 \alpha}$$

$$\times p^{d + c_1 - 1} (1 - p)^{r + c_2 - 1} t^{q + a_0 - 1} e^{-tH}$$

$$H = b_0 + pA(\alpha) + (1 - p)B(\alpha)$$

Hence $t \mid \alpha, p, D \sim \text{Gamma}(q + a_0, H)$. Integrating t exactly gives

$$\pi(\alpha, p \mid D) \propto \alpha^{q + a_1 - 1} e^{(\alpha - 1)C - b_1 \alpha}$$

$$\times p^{d + c_1 - 1} (1 - p)^{r + c_2 - 1} H^{-(q + a_0)}$$

The unrestricted support is $\alpha > 0$, $0 < p < 1$; the restricted support is $\alpha > 0$, $0 < p \leq 1/2$. Two-dimensional Gauss-Legendre quadrature integrates $\log \alpha$ from $\log(0.02)$ to $\log(30)$ and p on its model support. The $\exp(\log \alpha)$ Jacobian is included, and weights are normalized with log-sum-exp. Tail weights and refinement comparisons check truncation and discretization. Simulation uses 80 by 110 nodes, whereas the application uses 240 by 320 and a denser refinement. These numerical bounds are integration settings; edge-mass checks assess whether they materially truncate the posterior.

The conditional moments are

$$E(t \mid \alpha, p, D) = \frac{q + a_0}{H}$$

$$E(t^2 \mid \alpha, p, D) = \frac{(q + a_0)(q + a_0 + 1)}{H^2}$$

Consequently, the posterior rate means are

$$E(a \mid D) = E\left[\frac{p(q + a_0)}{H} \mid D\right]$$

$$E(b \mid D) = E\left[\frac{(1-p)(q + a_0)}{H} \mid D\right]$$

The posterior mean shape follows directly from the normalized quadrature weights.

Under squared error loss, the Bayes reliability estimate is the posterior mean of R itself:

$$\widehat{R}_B = E[R(p, 1-p) \mid D]$$

It is generally different from evaluating R at posterior mean rates. Under LINEX loss

$$L(\delta, R) = \exp[c(\delta - R)] - c(\delta - R) - 1$$

Differentiating posterior expected loss gives

$$\widehat{R}_{B,c} = -\tfrac{1}{c}\log E[e^{-cR} \mid D], \qquad c \neq 0$$

The notebook reports $c = +1.5$ and $c = -1.5$, equal-tail 95% credible intervals, and shortest weighted 95% intervals for R. Shortest intervals are numerical approximations to HPD intervals for a unimodal marginal; equal-tail and shortest intervals are not interchangeable.

For an independent computational check, four Metropolis-Hastings chains sample the collapsed density in $(\log \alpha, \operatorname{logit} v)$, where $p = v$ for the unrestricted model and $p = v/2$ for the restricted model. The transformed target includes the log-shape Jacobian and

$$\log\left(\frac{dp}{dz}\right) = \log p_{\max} + \log v + \log(1 - v)$$

Symmetric Gaussian random-walk proposals use a mode-based covariance fixed before sampling. Given each accepted state, t is drawn from its exact conditional Gamma distribution. Each chain uses 15,000 steps with 4,000 discarded warm-up steps; no adaptation occurs in the retained samples. Trace plots, density plots, split R-hat, effective sample sizes, and Monte Carlo standard errors accompany comparisons against quadrature. These diagnostics are classical split variance R-hat and an initial monotone autocorrelation ESS, not rank-normalized diagnostics.

9. Simulation design and measured performance

The factorial design comprises 3 order regimes x 2 shapes x 2 block sizes x 2 hybrid protocols x 2 record orders = 48 scenarios. Each has 300 independent replications, for 14,400 independently simulated datasets. There are three blocks, $n = 20$ or $n = 40$ per block, $m = \lfloor 0.7n \rfloor$, a balanced withdrawal plan, and T at the true strength distribution's 65th percentile. The threshold is fixed before generating each experiment. Shapes are 0.8 and 1.8; k is 1 or 3; the prescribed record count is 6 for $n = 20$ and 12 for $n = 40$. This size contrast increases both strength and stress information; it does not isolate the effect of strength sample size alone.

Rate regimes are ordered $(a, b) = (0.8, 1.6)$, boundary $(a, b) = (1, 1)$, and violated $(a, b) = (1.6, 0.8)$. Parameters are fixed, not sampled from priors. All four estimators use exactly the same simulated observations in a replication. Reported targets are α, a, b and $R_{3,5}$. Restricted estimators in the violated regime are intentionally misspecified and are judged against the true unrestricted target.

For estimates $\widehat{\theta}_l$ and truth θ, let $e_l = \widehat{\theta}_l - \theta$. Report

$$\text{Bias} = \frac{1}{M}\sum_{l=1}^{M} e_l, \qquad \text{MSE} = \frac{1}{M}\sum_{l=1}^{M} e_l^2$$

$$\text{RMSE} = \sqrt{\text{MSE}}, \qquad \text{MCSE(Bias)} = \frac{\text{SD}(e_l)}{\sqrt{M}}$$

$$\text{MCSE(MSE)} = \frac{\text{SD}(e_l^2)}{\sqrt{M}}$$

For intervals, report coverage $\widehat{\text{CP}}$, average length and

$$\text{MCSE}(\widehat{\text{CP}}) = \sqrt{\frac{\widehat{\text{CP}}(1 - \widehat{\text{CP}})}{M}}$$

The main coverage table includes unrestricted log-ratio asymptotic intervals and unrestricted/restricted Bayesian equal-tail intervals. Restricted MLE bootstrap coverage is separately measured in six selected scenarios with 100 outer replications and 199 inner resamples; its greater Monte Carlo uncertainty is displayed. The primary simulation's restricted MLE has no invented regular normal interval.

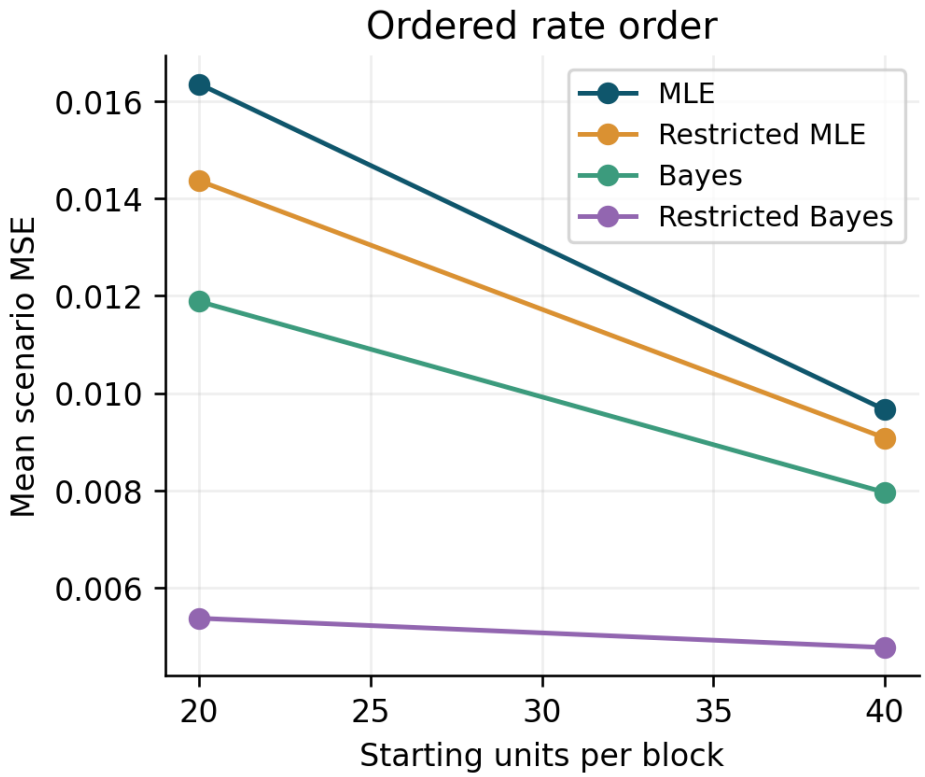
All raw replications, scenario settings, summary tables, numerical-refinement results and figures are exported. Seeds are fixed, and computational failures and specimen conservation are audited. Conclusions refer to the measured results rather than asserting universal Bayesian superiority.

Rate order	Estimator	Bias	MSE	Coverage
boundary	Bayes	0.0101	0.0115	0.9625
boundary	MLE	0.0057	0.0171	0.9379
boundary	Restricted Bayes	0.1076	0.0142	0.0000
boundary	Restricted MLE	0.0550	0.0097	n/a
ordered	Bayes	-0.0283	0.0099	0.9673
ordered	MLE	-0.0020	0.0130	0.9410
ordered	Restricted Bayes	-0.0044	0.0051	0.9835
ordered	Restricted MLE	0.0004	0.0117	n/a
violated	Bayes	0.0345	0.0102	0.9587
violated	MLE	0.0061	0.0126	0.9400
violated	Restricted Bayes	0.2501	0.0632	0.0000
violated	Restricted MLE	0.1969	0.0393	n/a

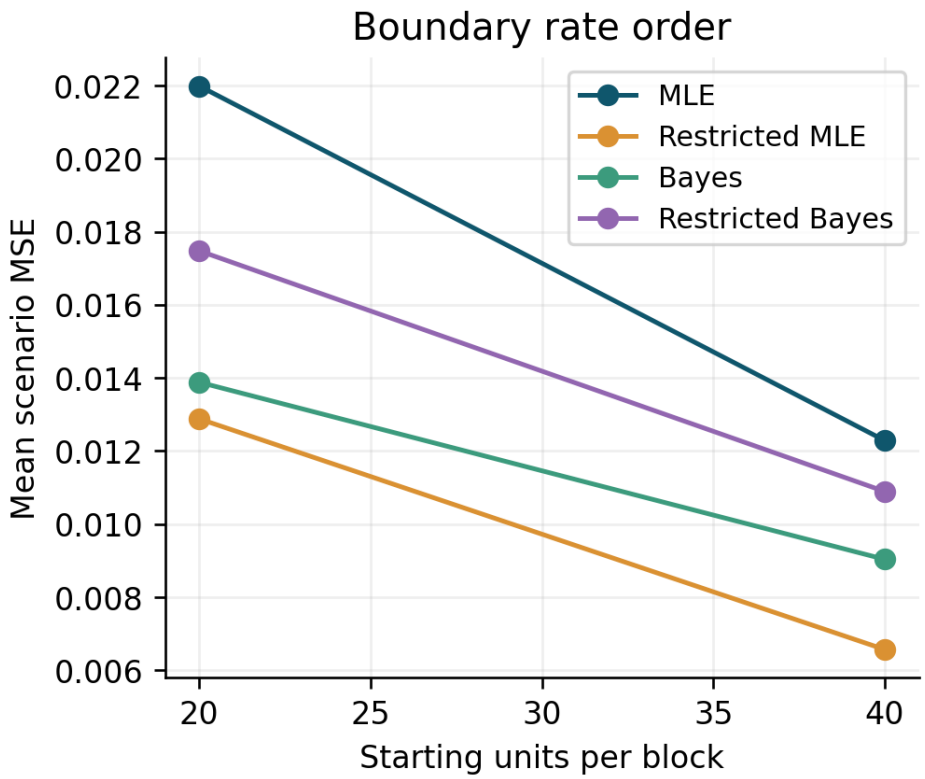
This table averages scenario-specific metrics within each regime. Restricted MLE confidence coverage is not included here because the main simulation does not assign it a regular boundary-normal interval. Its bootstrap assessment follows separately. Scenario-level values and their MCSEs are available in simulation_summary.csv.

10. Simulation figures and restricted bootstrap coverage

Measured scenario-averaged reliability bias and MSE. Each plotted size value averages eight scenarios.

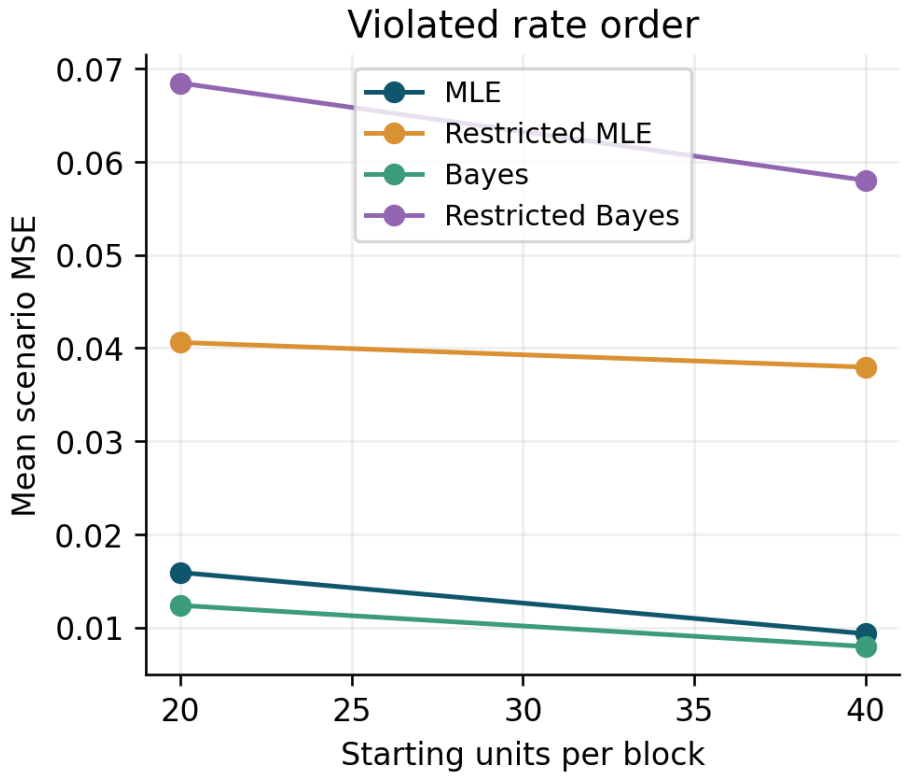


Ordered rate order

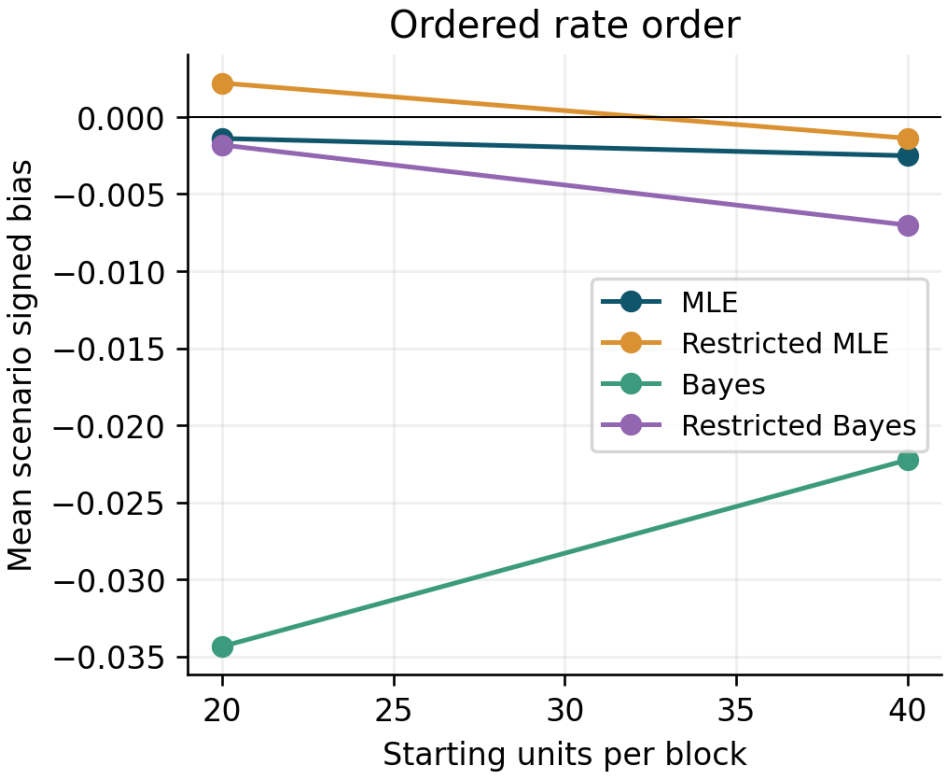


Boundary rate order

Measured scenario-averaged reliability bias and MSE. Each plotted size value averages eight scenarios.

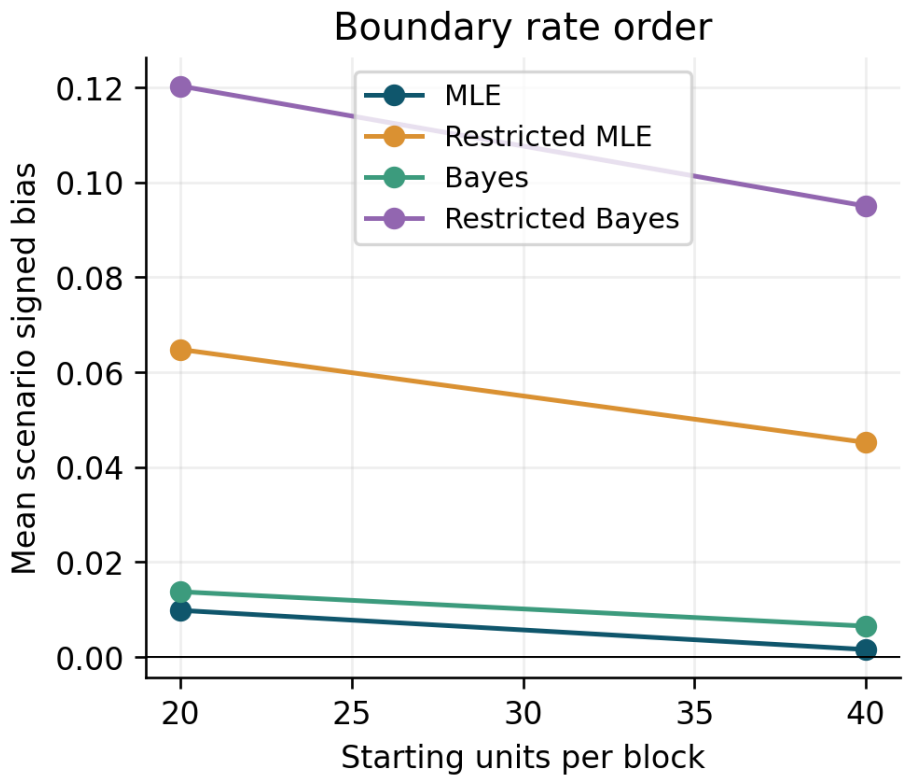


Violated rate order

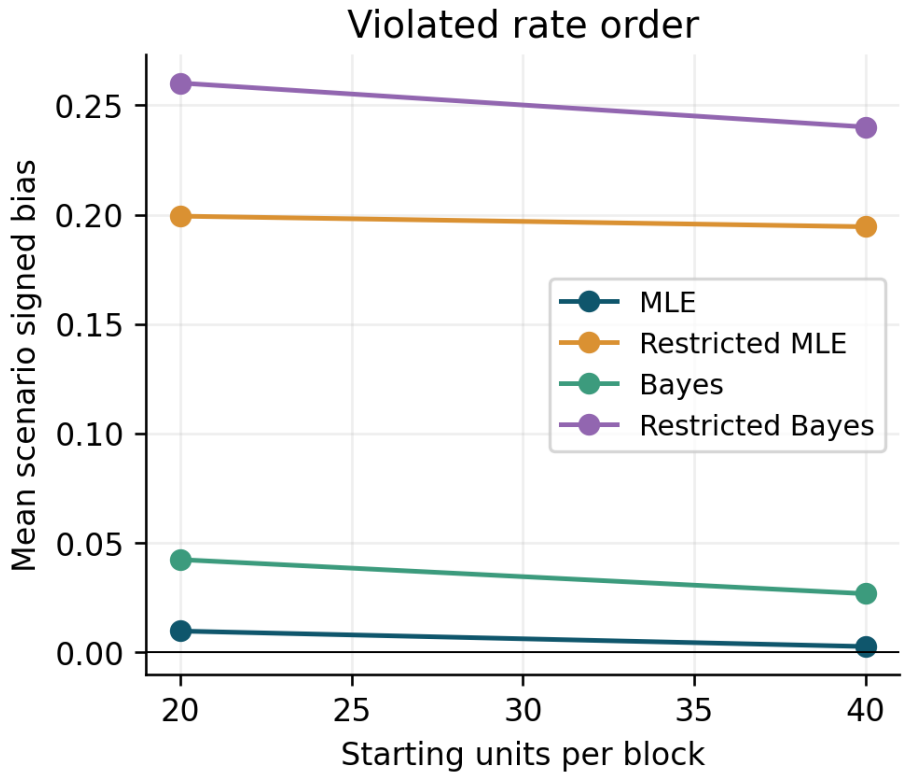


Ordered rate order

Measured scenario-averaged reliability bias and MSE. Each plotted size value averages eight scenarios.

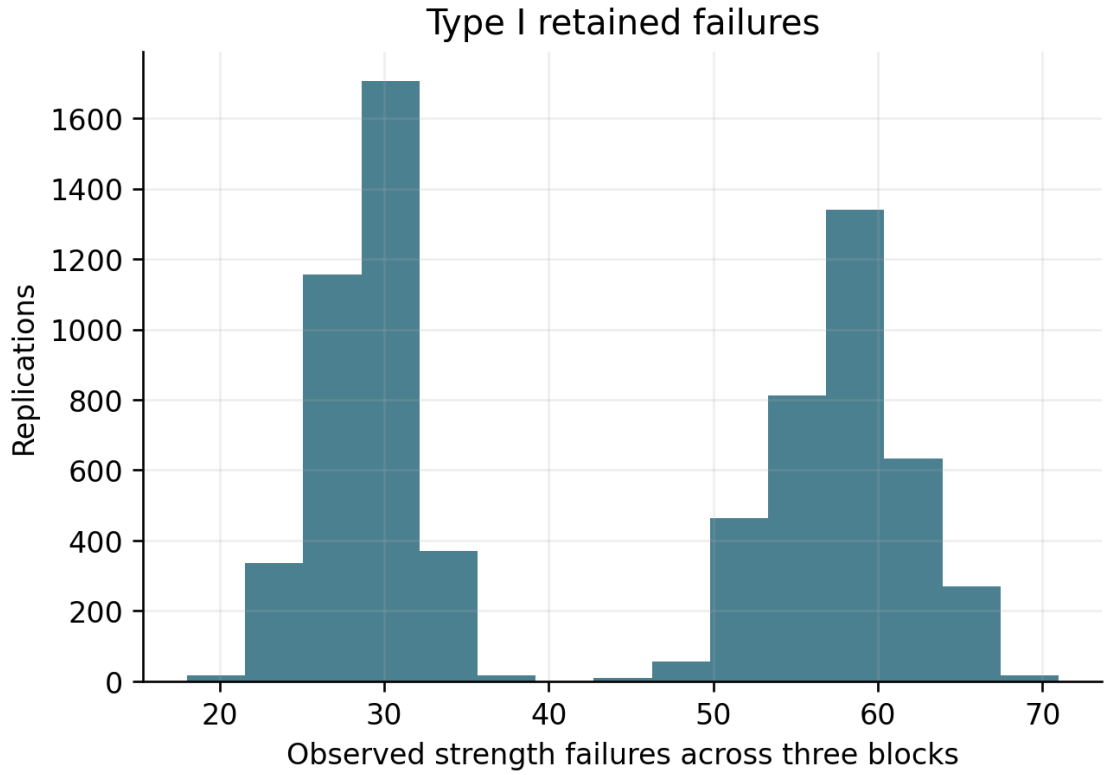


Boundary rate order

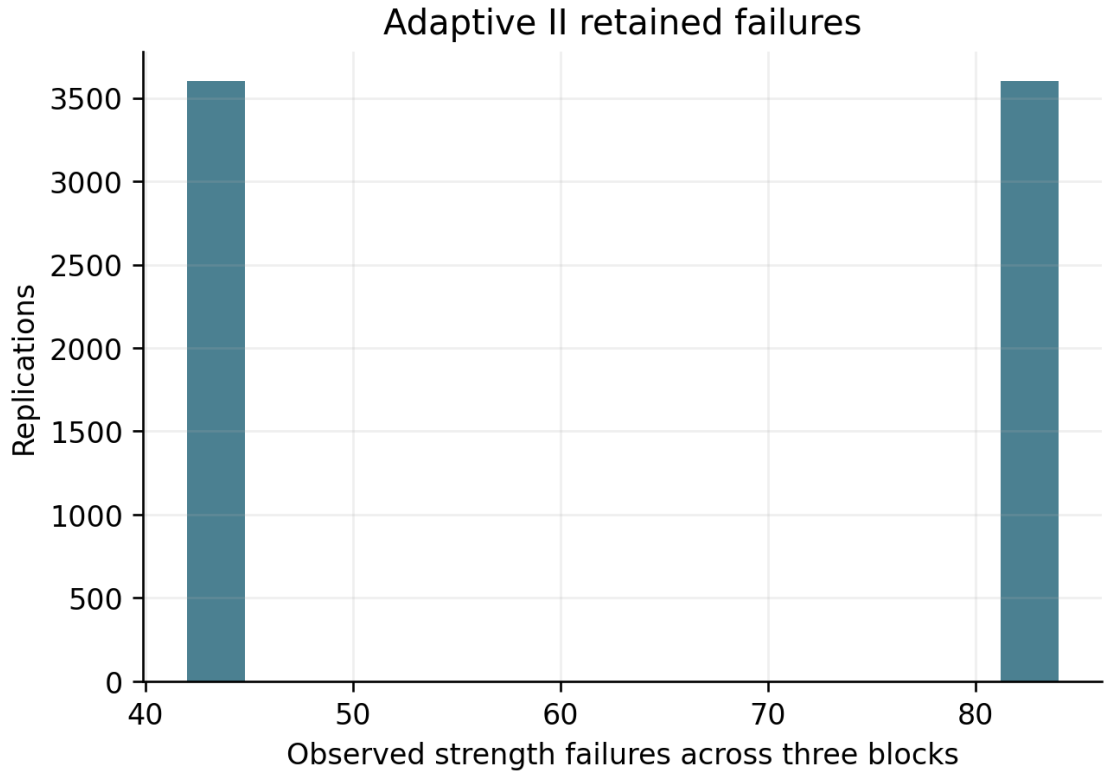


Violated rate order

Retained strength-failure counts under both hybrid protocols

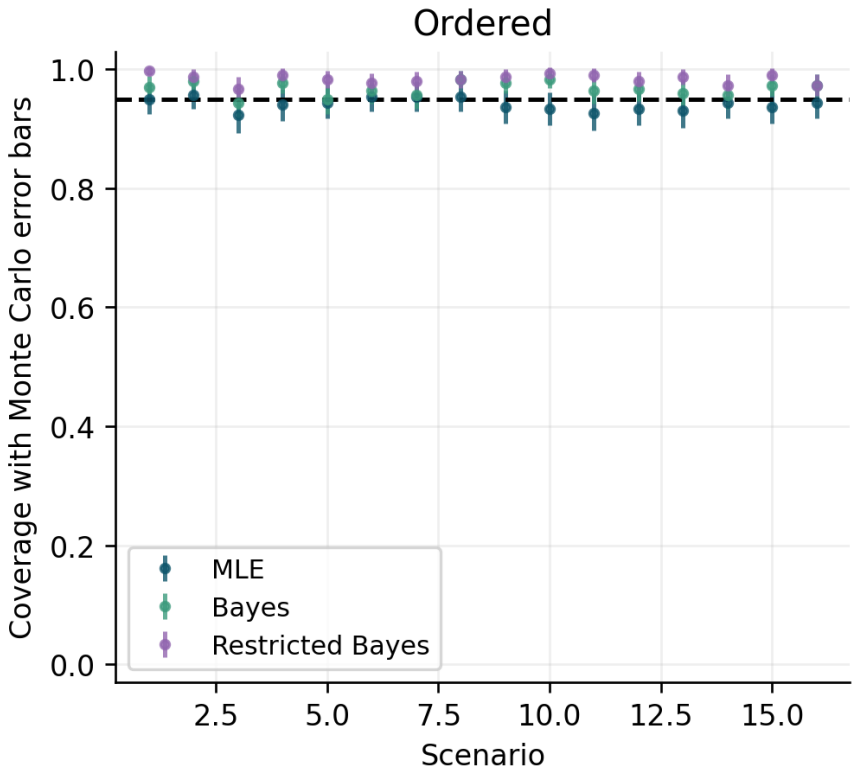


Type I retained failures

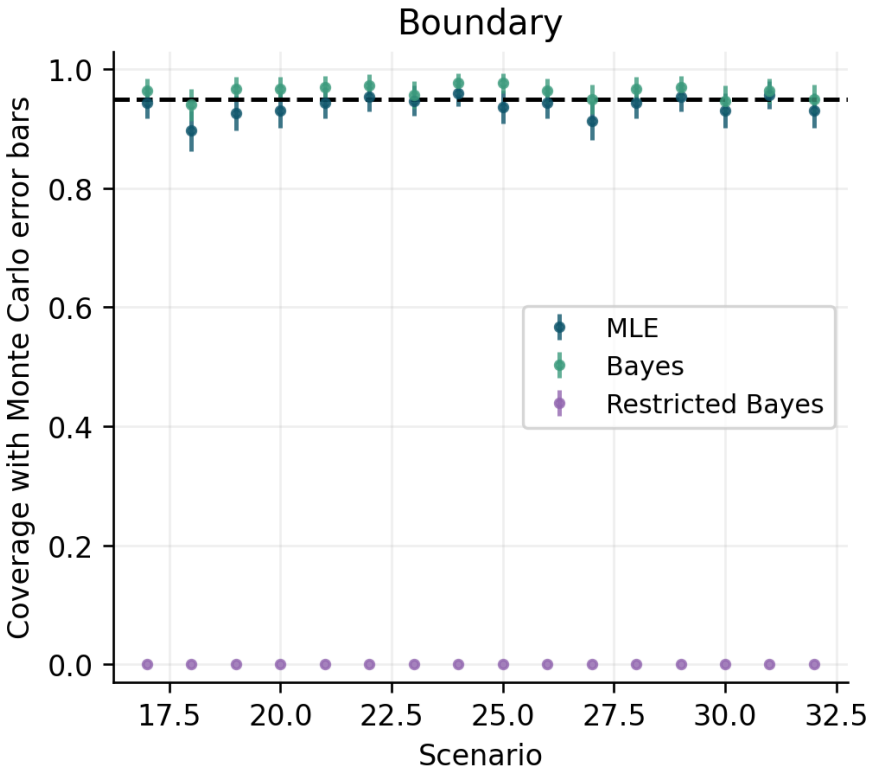


Adaptive II retained failures

Scenario-level coverage with 1.96 times Monte Carlo standard-error bars.

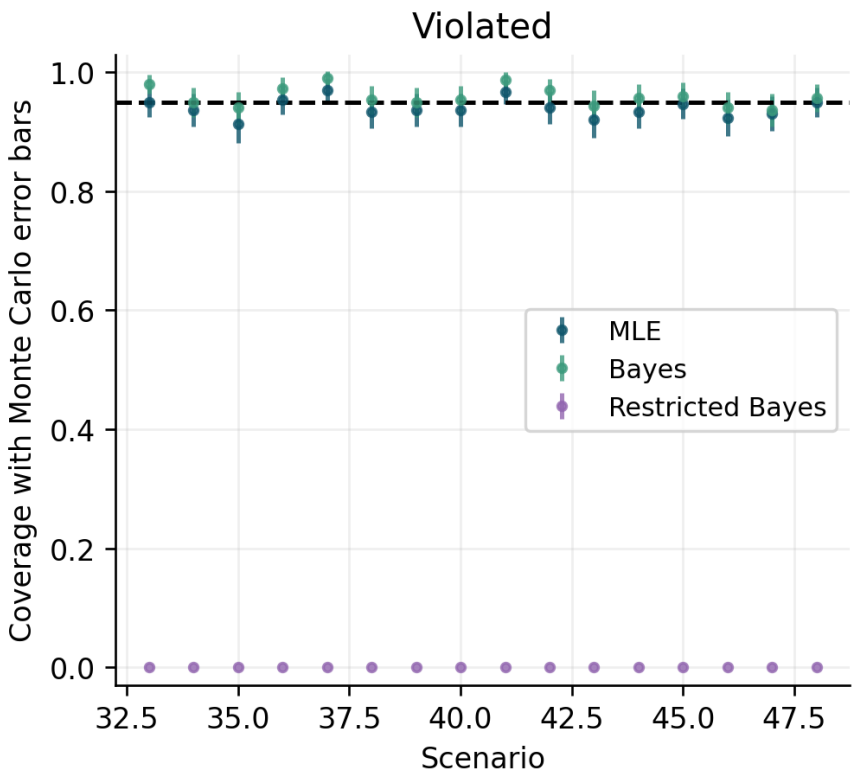


Ordered



Boundary

Scenario-level coverage with 1.96 times Monte Carlo standard-error bars.



Violated

Rate order	Protocol	Coverage	Coverage MC error	Mean length
ordered	Type I	0.9100	0.0286	0.3706
ordered	Adaptive II	0.8600	0.0347	0.3639
boundary	Type I	0.9400	0.0237	0.3403
boundary	Adaptive II	0.9100	0.0286	0.3434
violated	Type I	0.0000	0.0000	0.3210
violated	Adaptive II	0.0000	0.0000	0.3056

Restricted bootstrap coverage uses six selected scenarios, 100 outer datasets each and 199 inner resamples per dataset. Its Monte Carlo precision is lower than the primary simulation. False-order reliability truths below one half cannot be covered by intervals restricted above one half. Continuous ordered Bayesian equal-tail intervals also exclude the equality endpoint, explaining their boundary coverage behavior. No universal advantage is inferred from narrower intervals alone.

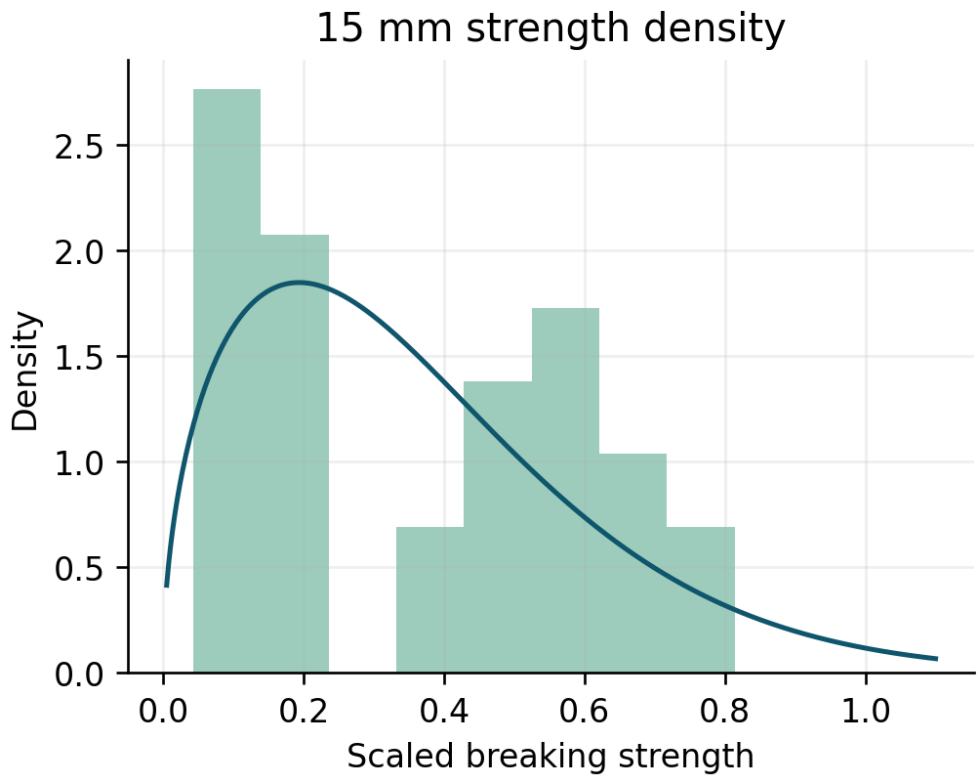
11. Jute-fibre data provenance and fit checks

Data are transcribed from Sultana et al., Table 10, printed page 3453 (PDF page 19), attributed to Xia et al. (2009). Each of the 15 mm and 20 mm gauge groups has 30 breaking-strength measurements already divided by 1000. The 15 mm group represents strength and the 20 mm group a same-unit laboratory comparison stress. The printed 70.32383 anomaly in the 10 mm group is visually confirmed; that group is excluded rather than silently repaired.

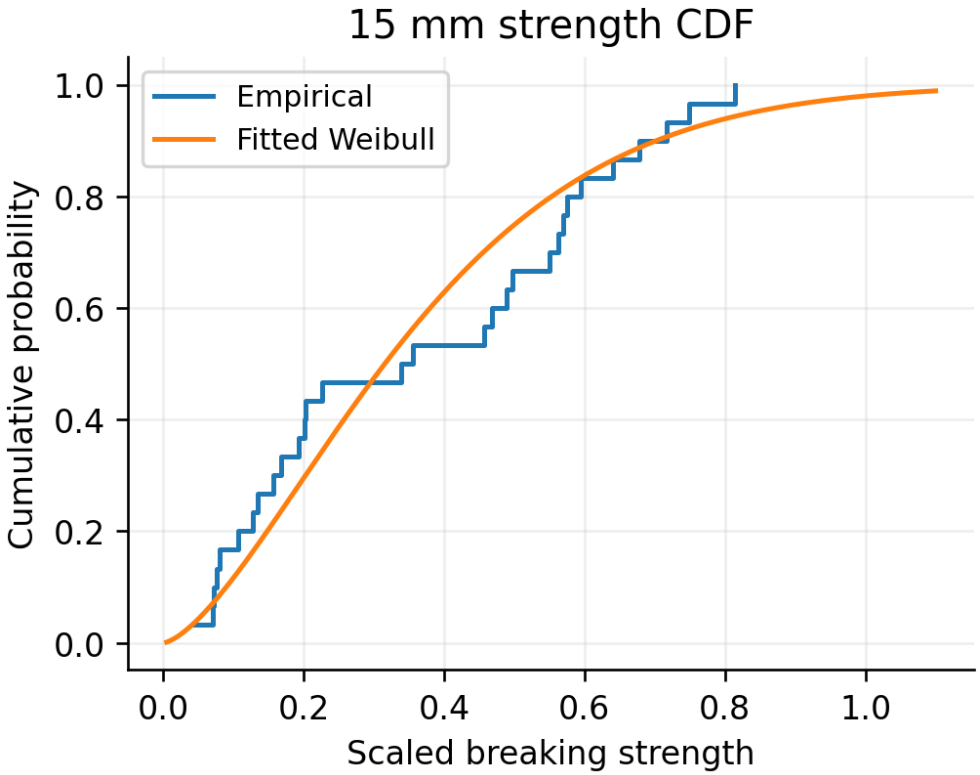
Sample	Size	Shape α	Rate	K-S	Bootstrap p
15 mm	30	1.4991	3.9080	0.1688	0.0260
20 mm	30	1.3607	3.8392	0.1490	0.0779

The two-population common-shape likelihood-ratio statistic is 0.2083, with one degree of freedom and bootstrap p-value 0.6204. Each goodness-of-fit calibration and the common-shape calibration uses 1,000 resamples with refitting. The 15 mm K-S result rejects Weibull adequacy at 5%, so its reliability estimates carry an explicit model-fit limitation. A high uncalibrated fitted-parameter K-S p-value would not resolve this issue.

Complete jute measurements: fitted density, empirical/fitted CDF and Q-Q diagnostics.

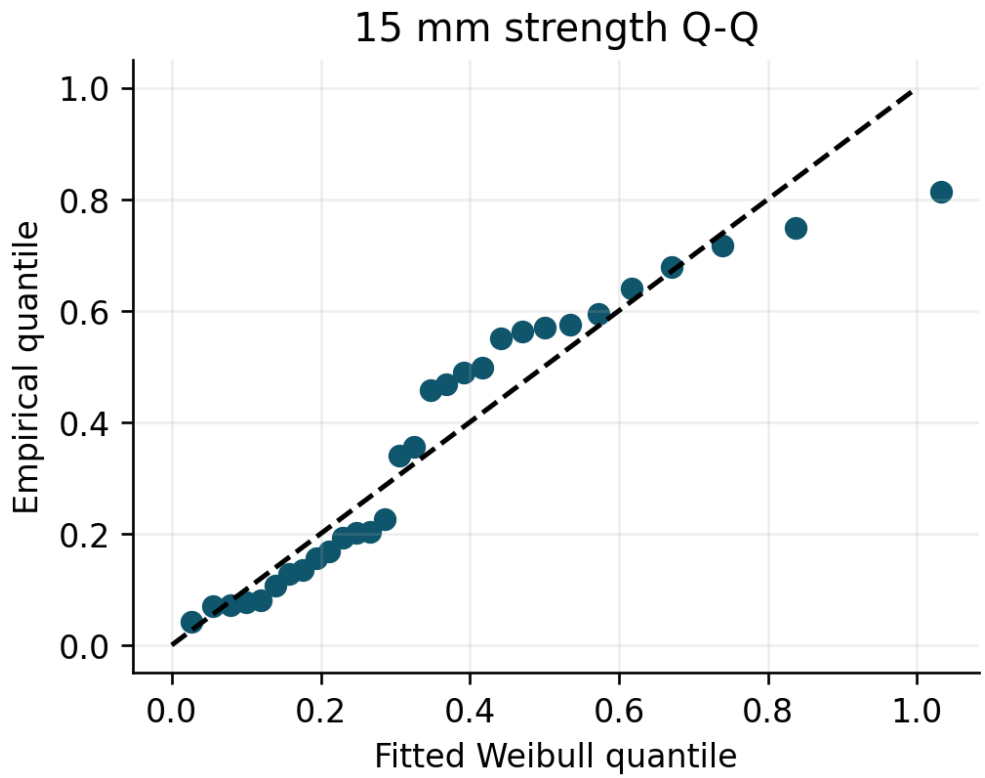


15 mm strength density

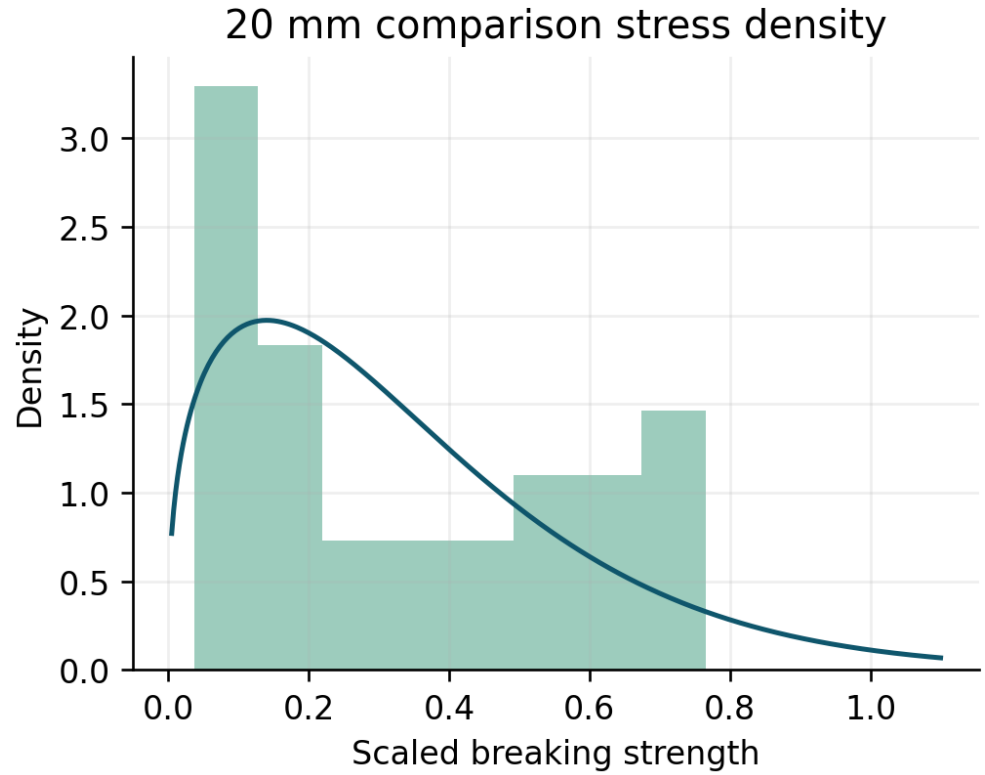


15 mm strength CDF

Complete jute measurements: fitted density, empirical/fitted CDF and Q-Q diagnostics.

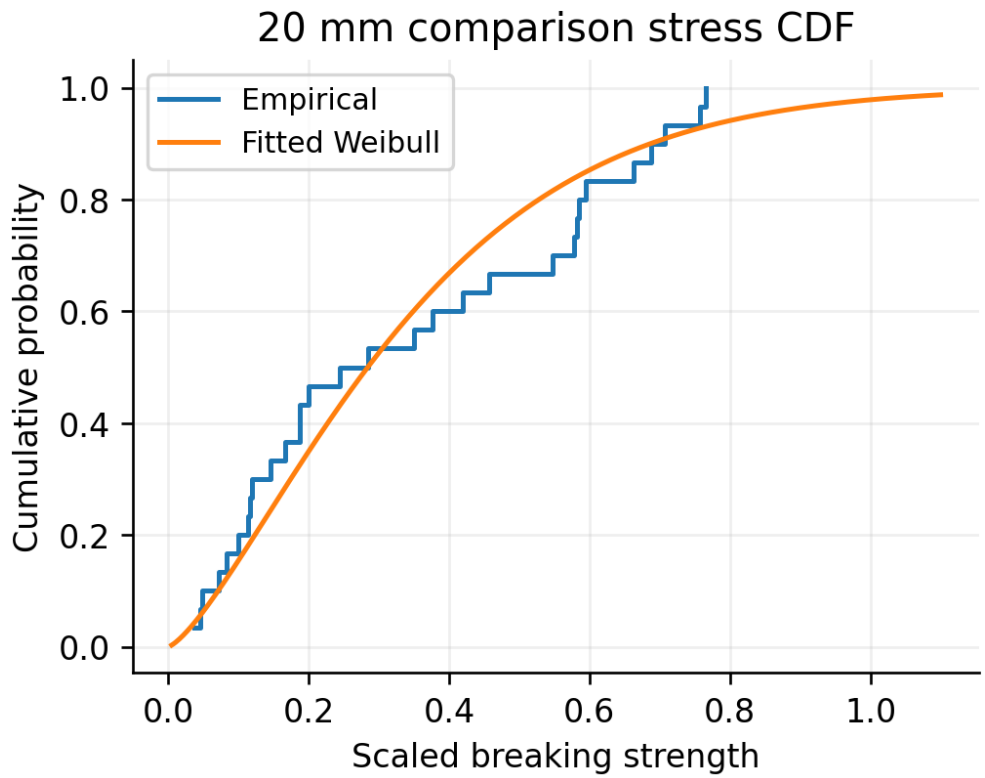


15 mm strength Q-Q

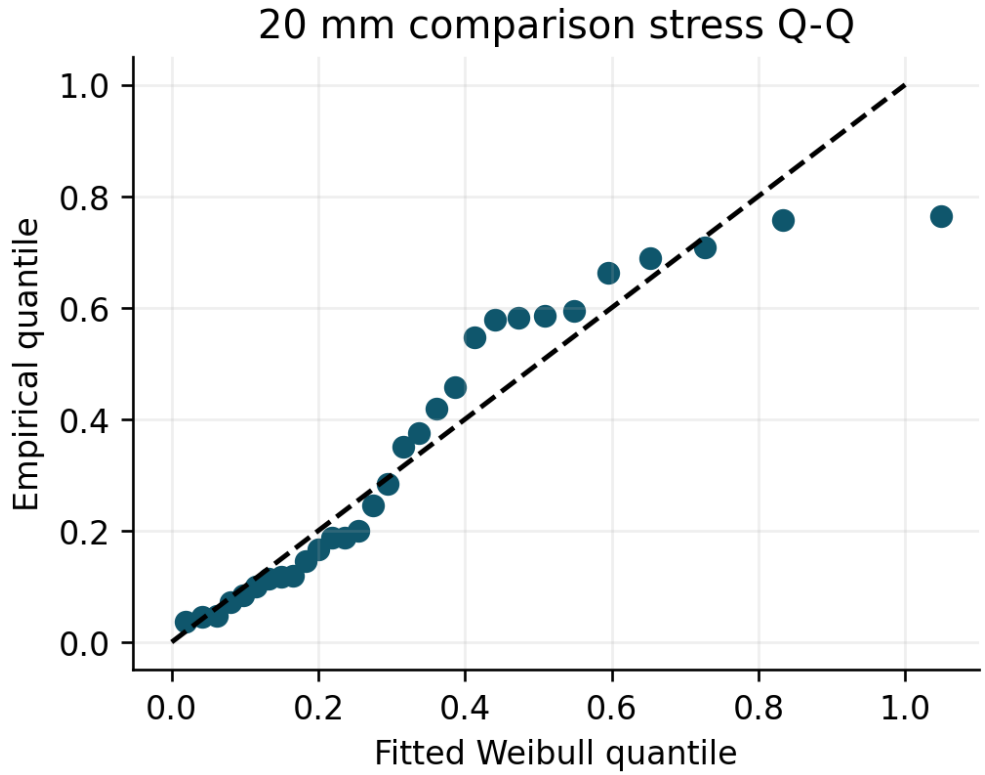


20 mm comparison stress density

Complete jute measurements: fitted density, empirical/fitted CDF and Q-Q diagnostics.



20 mm comparison stress CDF

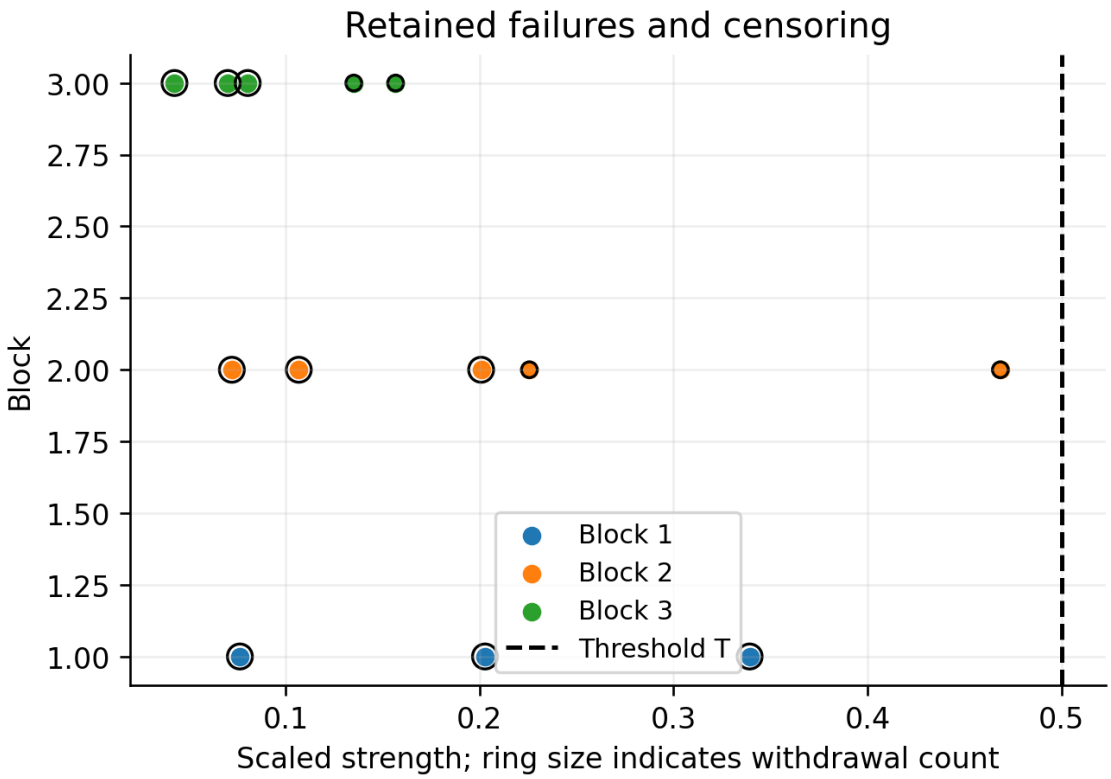


20 mm comparison stress Q-Q

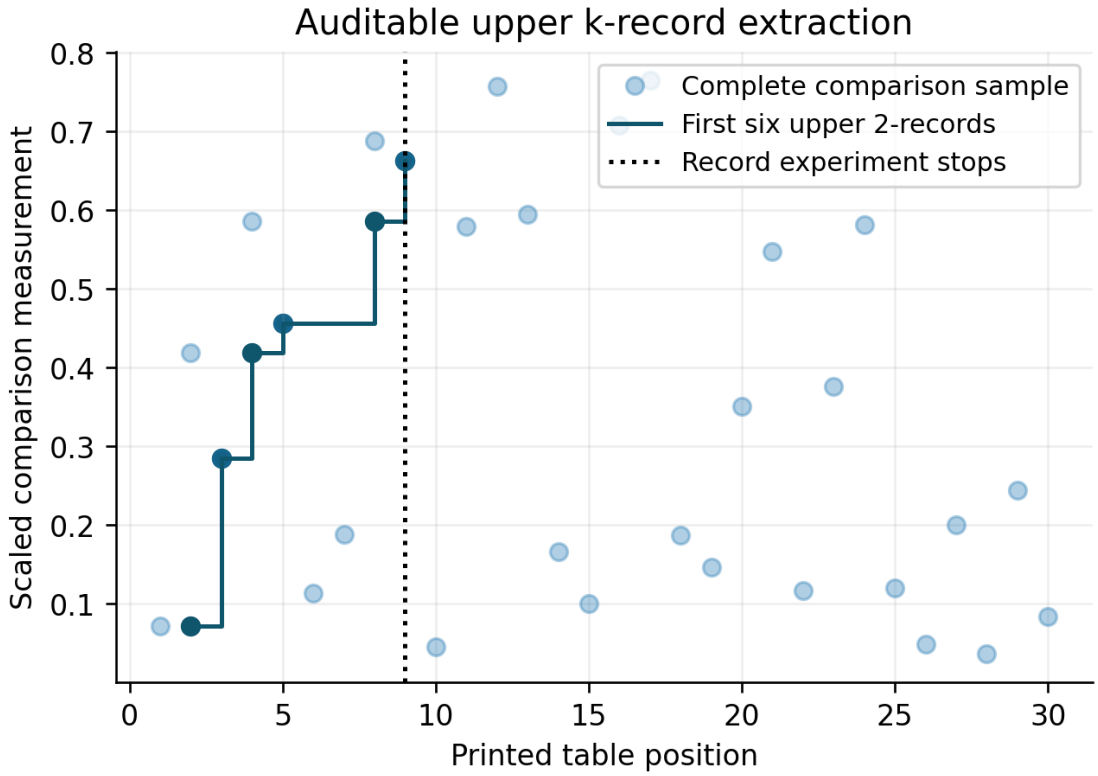
12. Jute retained observations and reliability estimates

A seeded allocation splits the strength group into three blocks of ten. Each uses target seven, T=0.5 and balanced withdrawals. The comparison group supplies the first six upper 2-records in printed table order, stopping at position nine. The printed order is an imposed acquisition order, not a historical record sequence. Complete measurements are used for fit checks; the censored/record likelihood uses only the retained experiment.

Strength failures with withdrawals and auditable upper-record extraction.



Retained failures and censoring

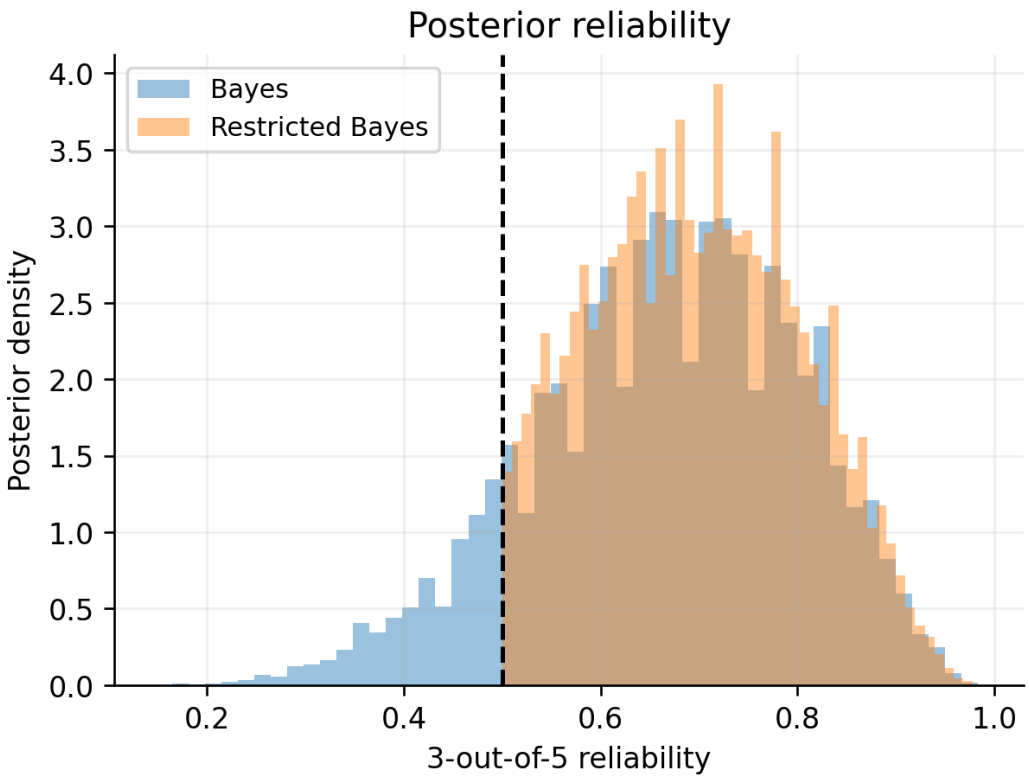


Auditable upper k-record extraction

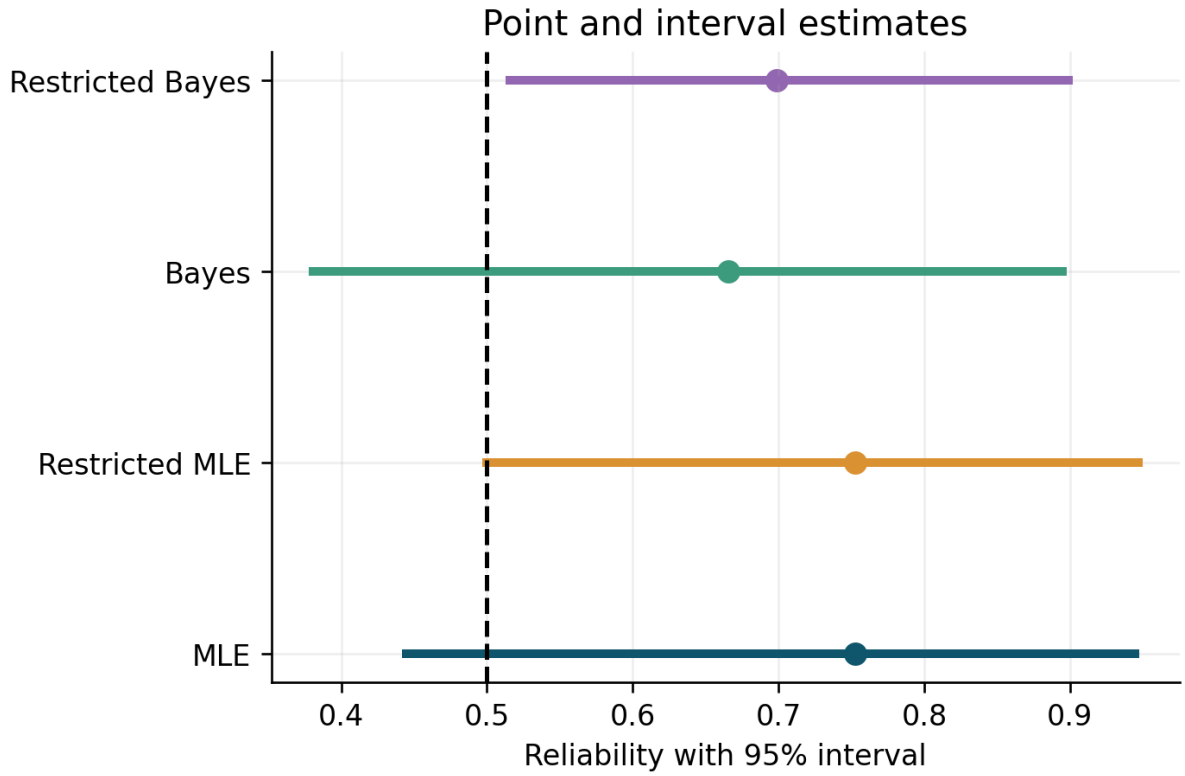
Estimator	Shape α	Strength rate a	Stress rate b	Reliability	95% lower	95% upper
MLE	1.1656	2.1167	4.8465	0.7526	0.4445	0.9446
Restricted MLE	1.1656	2.1167	4.8465	0.7526	0.5000	0.9469
Bayes	1.1788	2.2013	3.9011	0.6656	0.3805	0.8947
Restricted Bayes	1.1654	2.1039	4.1271	0.6989	0.5157	0.8988

MLE intervals use the unrestricted log-ratio approximation; restricted MLE intervals are parametric bootstrap percentile intervals. Bayesian rows use equal-tail credible intervals. Parameter intervals, LINEX values, shortest intervals and both bootstrap samples are retained in the executed notebook and exported result tables.

Actual posterior reliability densities and point/interval comparisons.



Posterior reliability



Point and interval estimates

13. Additional carbon-fibre validation

The supplied Newer and Alanazi article lists 100 carbon-fibre strengths in GPa on PDF pages 12 and 14, attributing them to Nichols and Padgett (2006). A seeded random split creates disjoint groups of fifty for strength and comparison stress. Both groups measure breaking strengths; no river discharge data are used. A fixed unit of 3 GPa scales measurements before inference. The same fixed prior is used on the scaled rates.

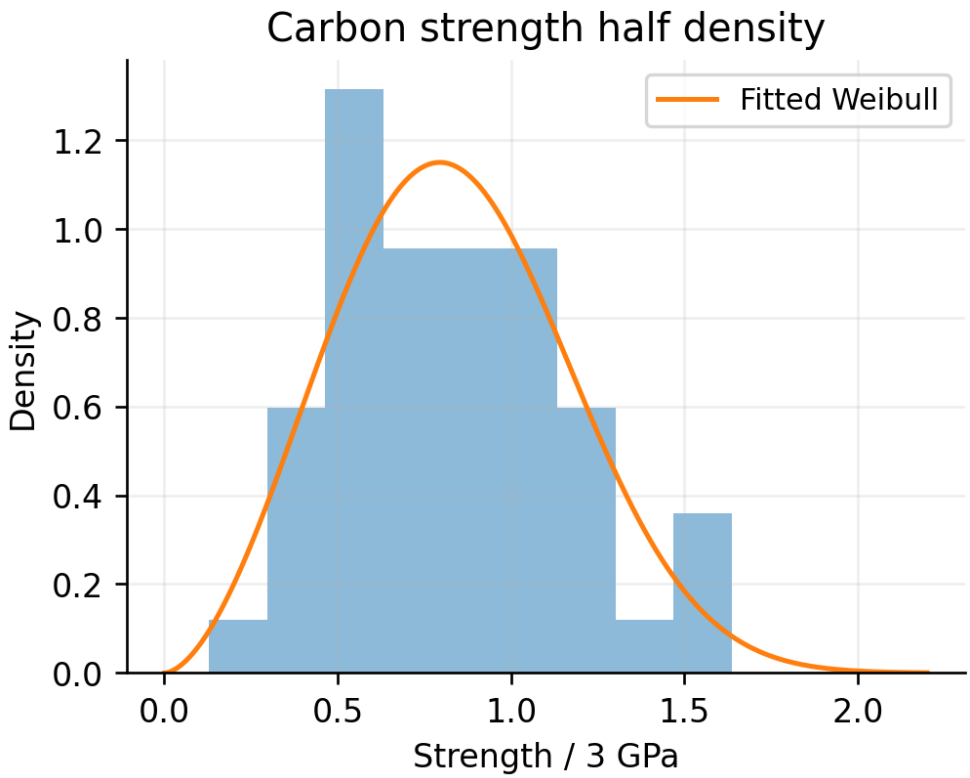
Sample	Size	Shape α	Rate	K-S	Bootstrap p
Carbon strength half	50	2.7186	1.1801	0.0776	0.5854
Carbon comparison half	50	2.8973	0.9421	0.1132	0.1009

Both calibrated K-S tests are compatible with Weibull at 5%. The common-shape bootstrap p-value is 0.6783. This provides a suitable supplementary empirical validation, while retaining the jute fit limitation. Compatibility is not proof of the distribution.

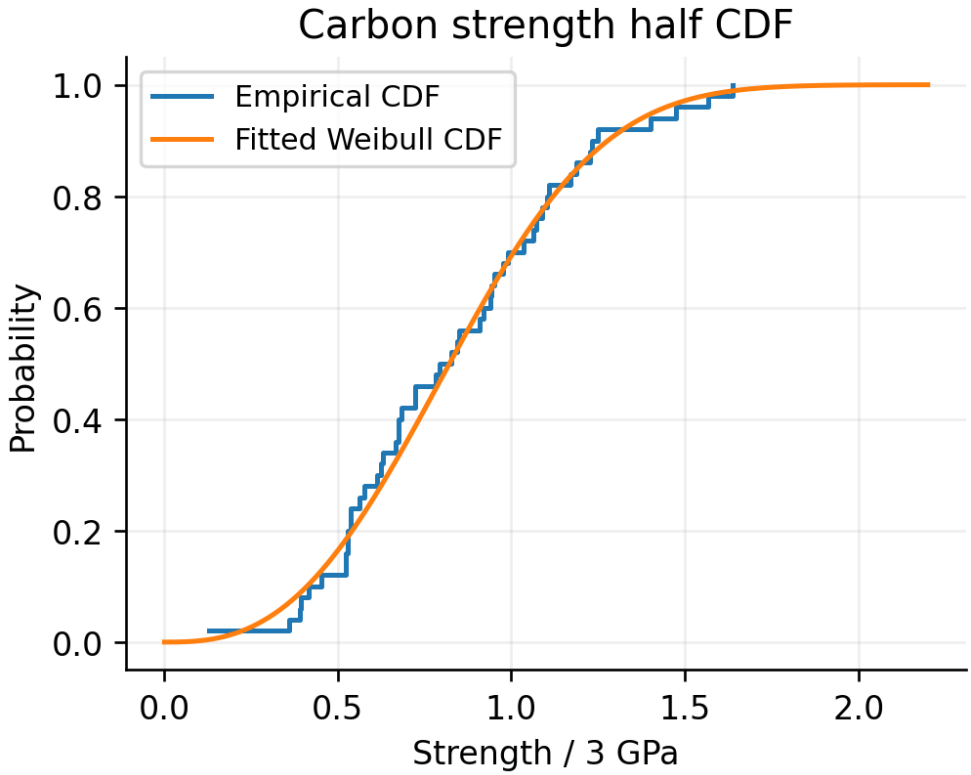
Strength blocks have 17,17 and 16 specimens, target floor(0.7n), T=1 in scaled units and balanced withdrawals. The comparison half supplies the first five upper 4-records. Retained record values are distinct; no jitter is applied to the rounded source data. The allocation, records, calibrated tests and all four estimates are saved.

Estimator	Shape α	Strength rate a	Stress rate b	Reliability	95% lower	95% upper
MLE	2.8742	1.3406	0.6949	0.3160	0.1327	0.6146
Restricted MLE	2.6605	1.1351	1.1351	0.5000	0.5000	0.8623
Bayes	2.7836	1.3169	0.8269	0.3623	0.1588	0.6081
Restricted Bayes	2.5523	1.0802	1.3382	0.5653	0.5020	0.7039

Additional carbon-fibre Weibull diagnostics in the fixed 3 GPa unit.

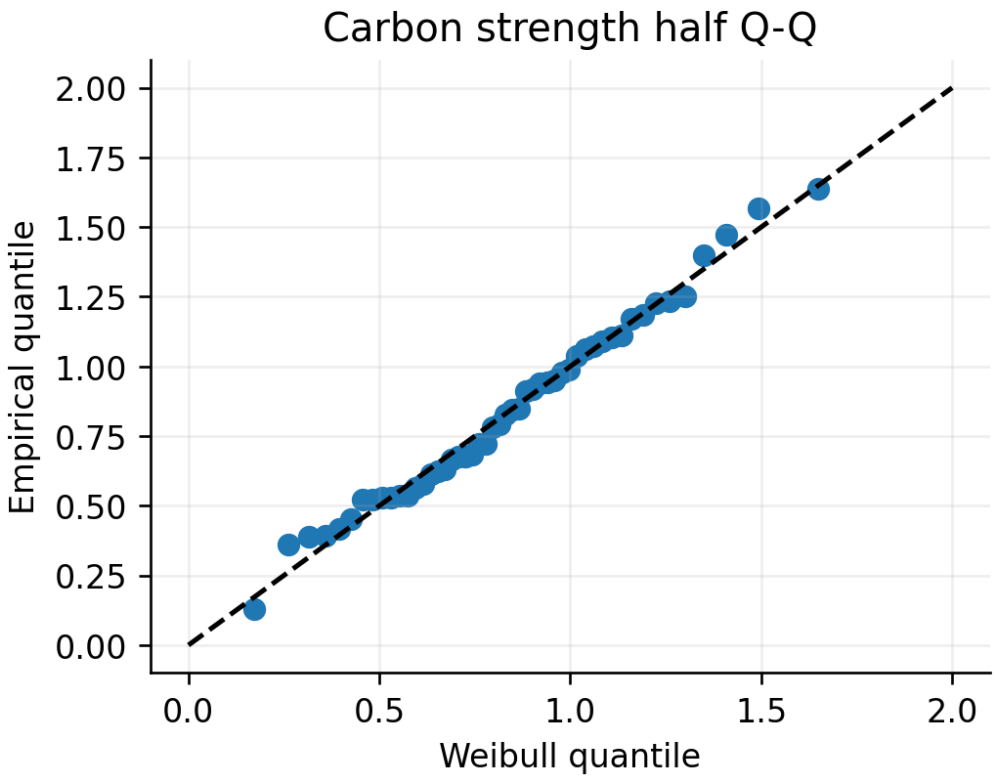


Carbon strength half density

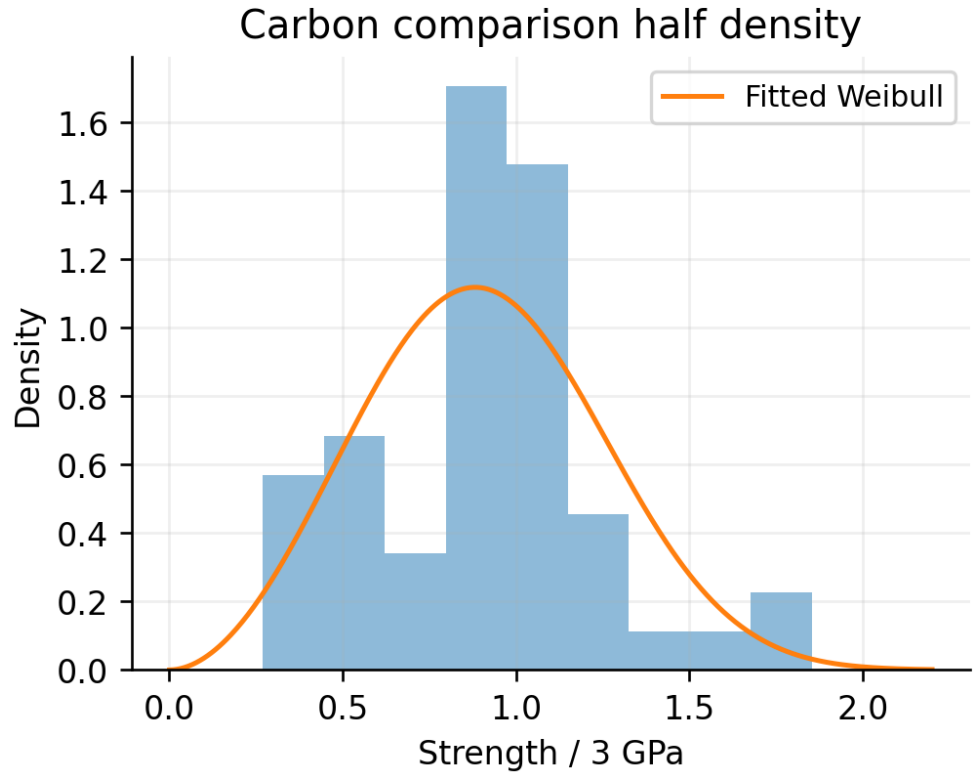


Carbon strength half CDF

Additional carbon-fibre Weibull diagnostics in the fixed 3 GPa unit.

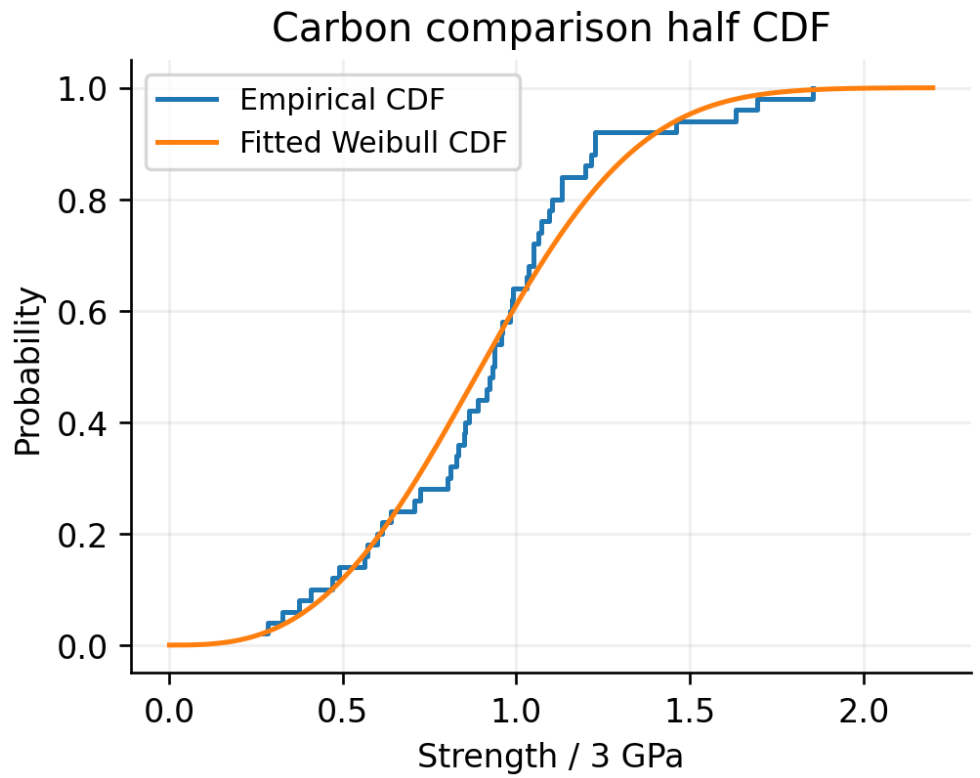


Carbon strength half Q-Q

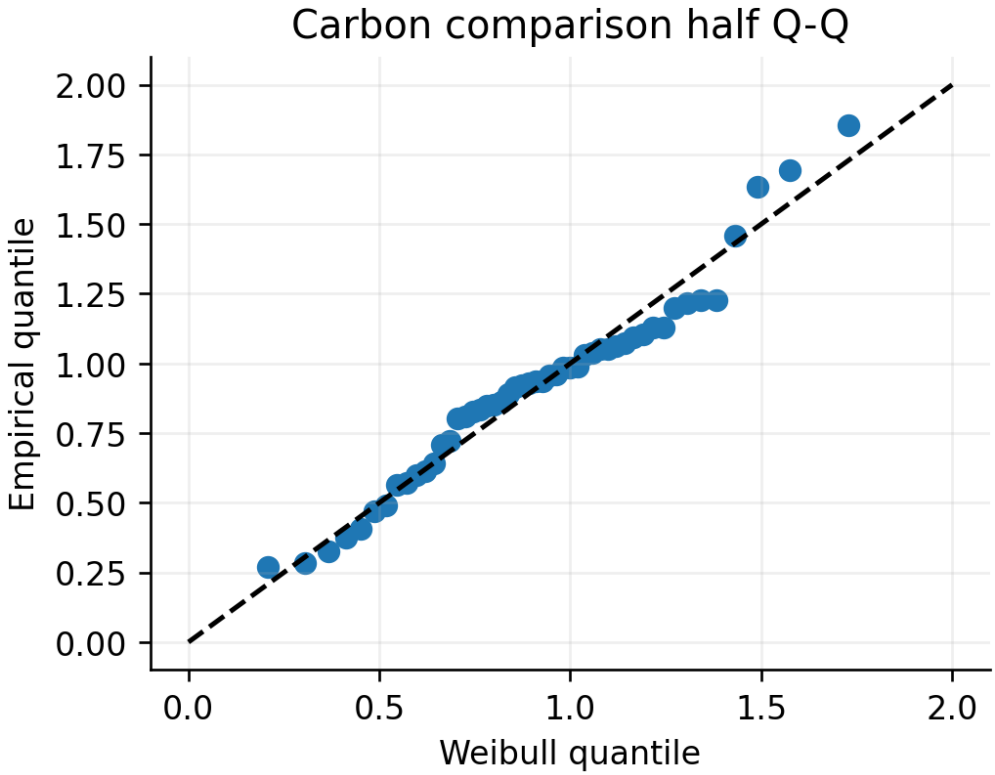


Carbon comparison half density

Additional carbon-fibre Weibull diagnostics in the fixed 3 GPa unit.



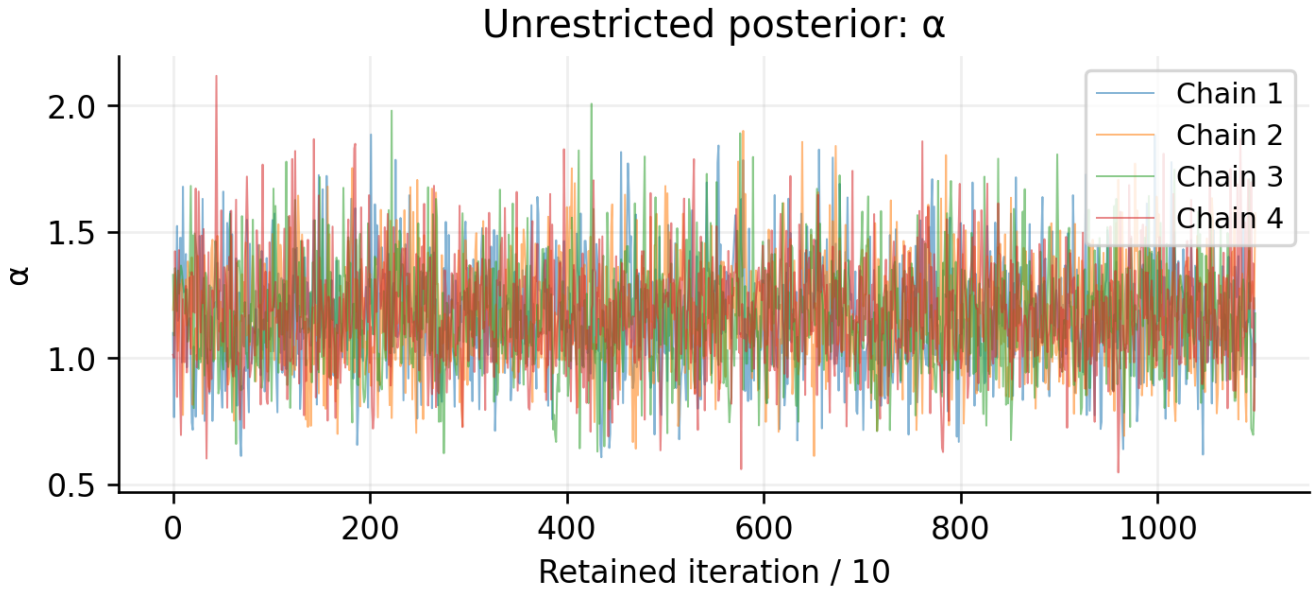
Carbon comparison half CDF



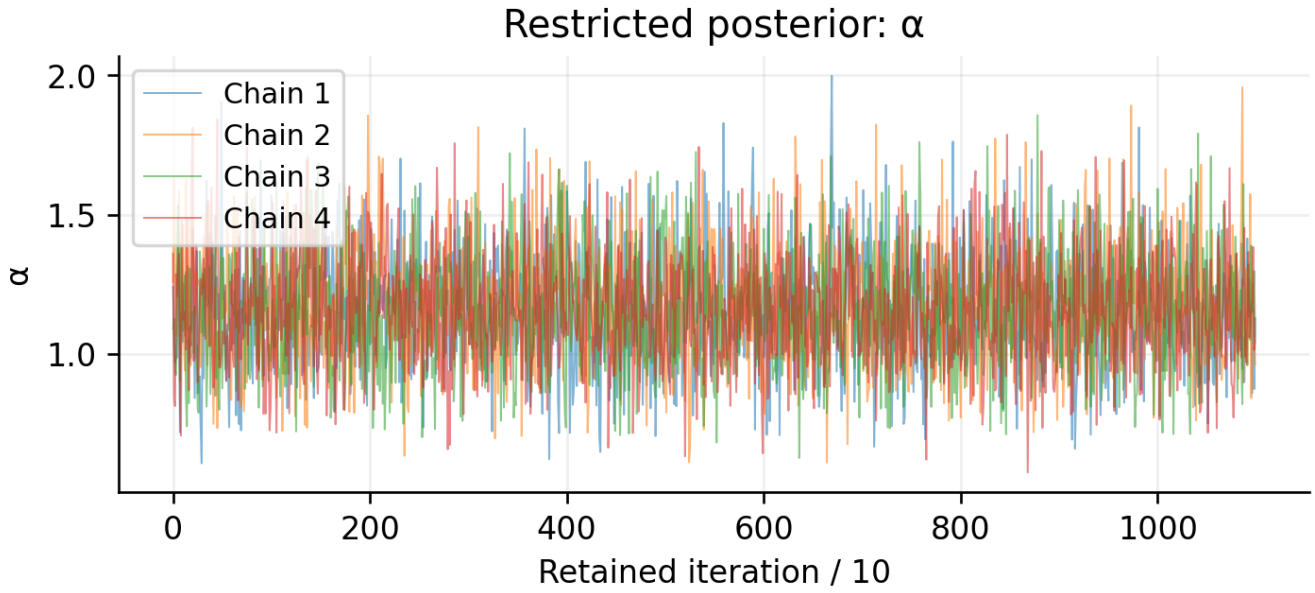
Carbon comparison half Q-Q

14. Posterior trace and density figures

Four actual retained chains for each parameter and reliability, shown every tenth iteration for readability.

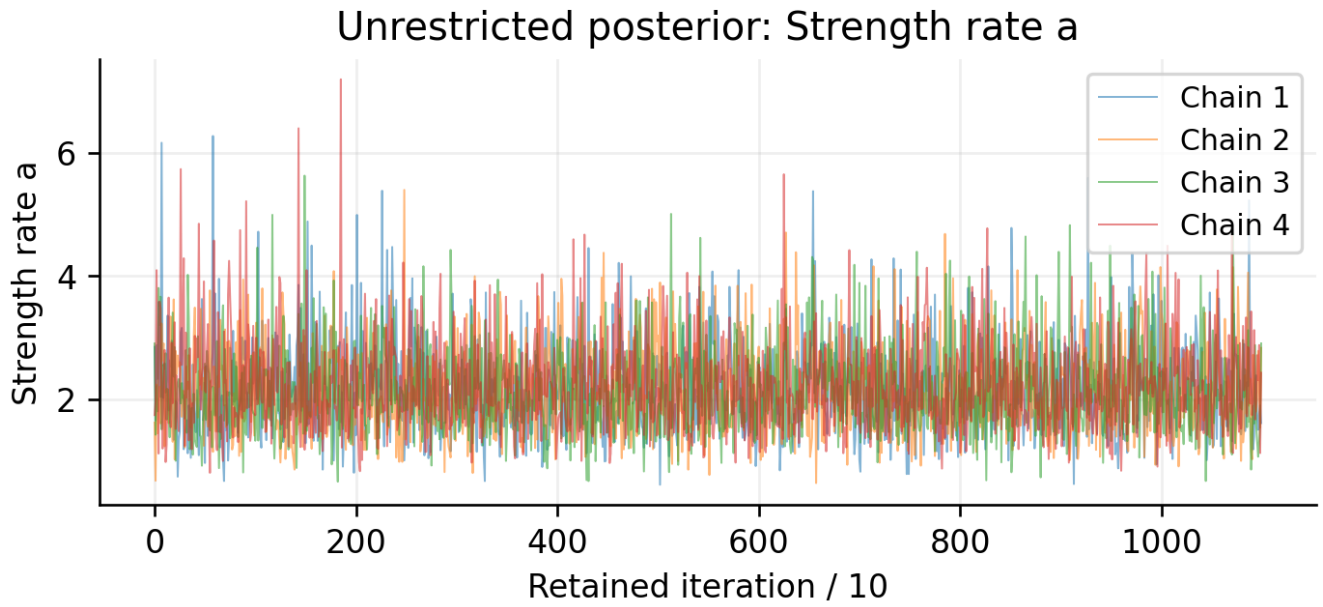


Unrestricted posterior: α

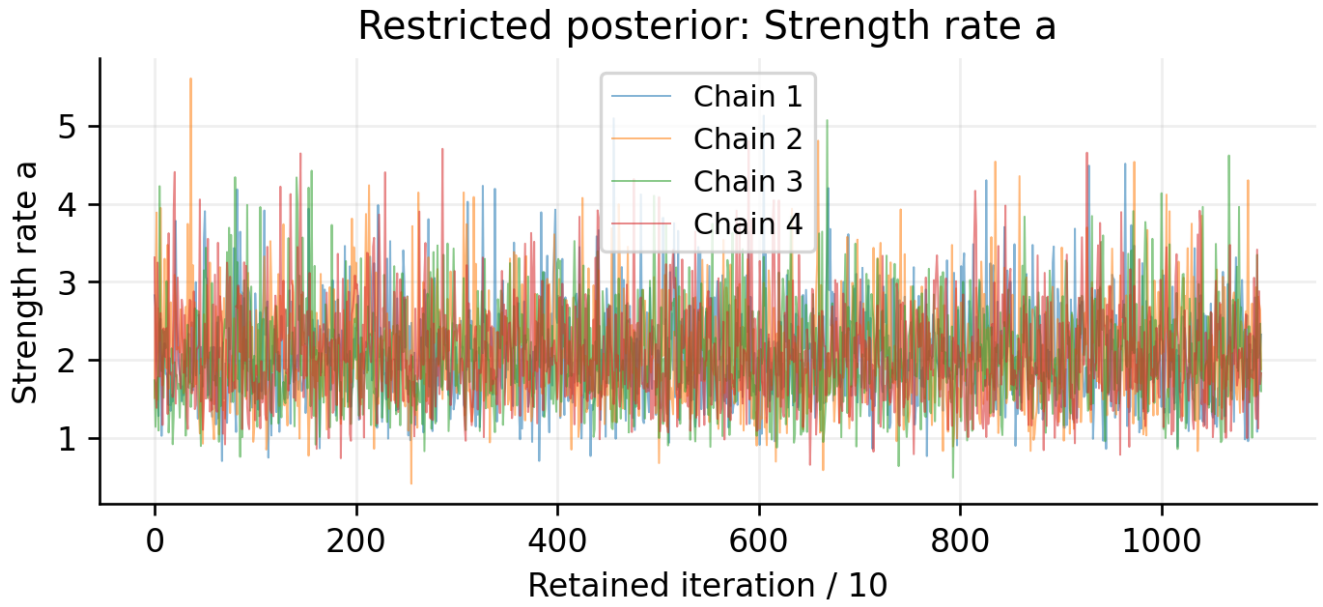


Restricted posterior: α

Four actual retained chains for each parameter and reliability, shown every tenth iteration for readability.

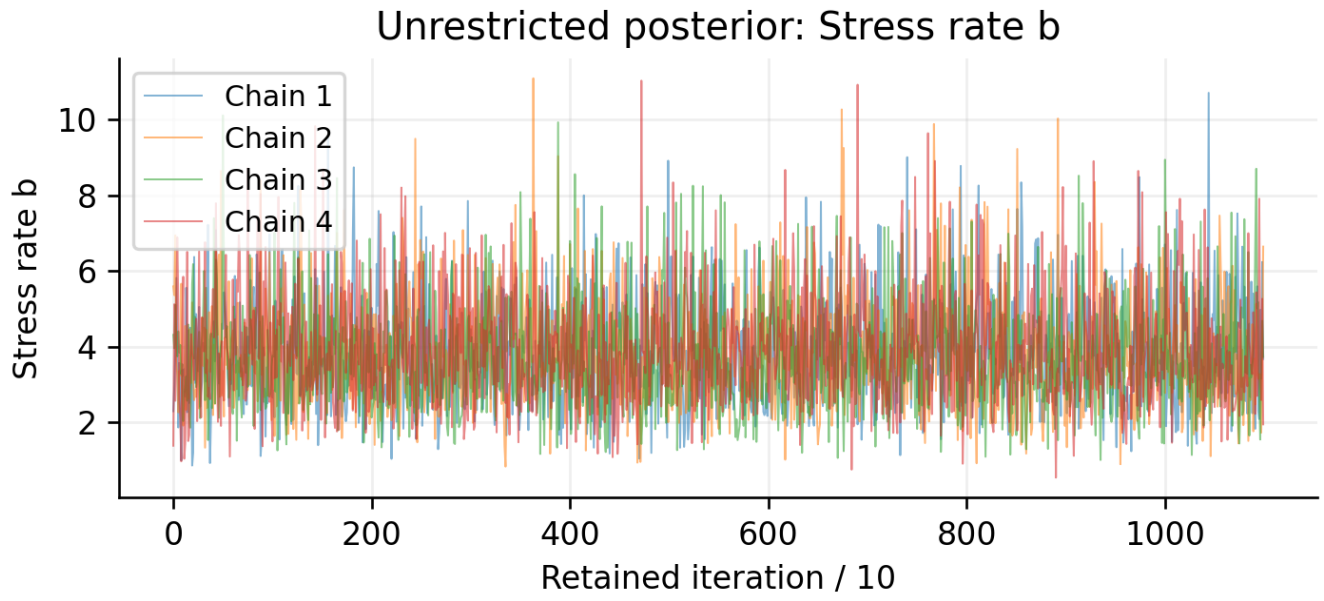


Unrestricted posterior: Strength rate a

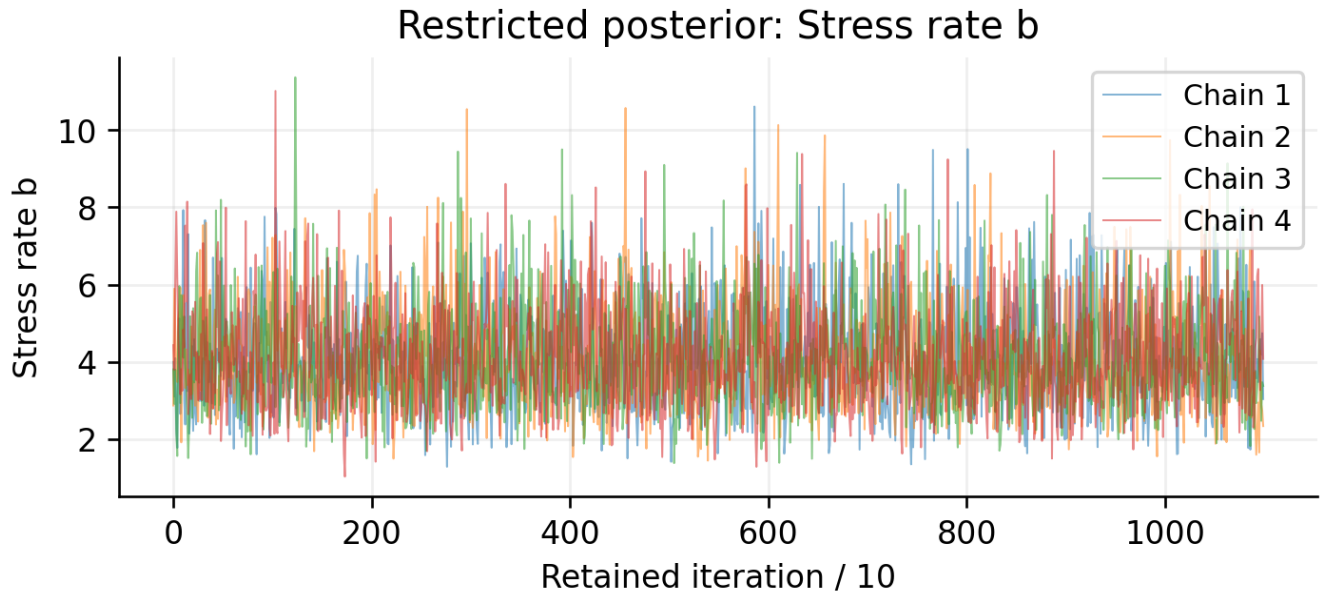


Restricted posterior: Strength rate a

Four actual retained chains for each parameter and reliability, shown every tenth iteration for readability.

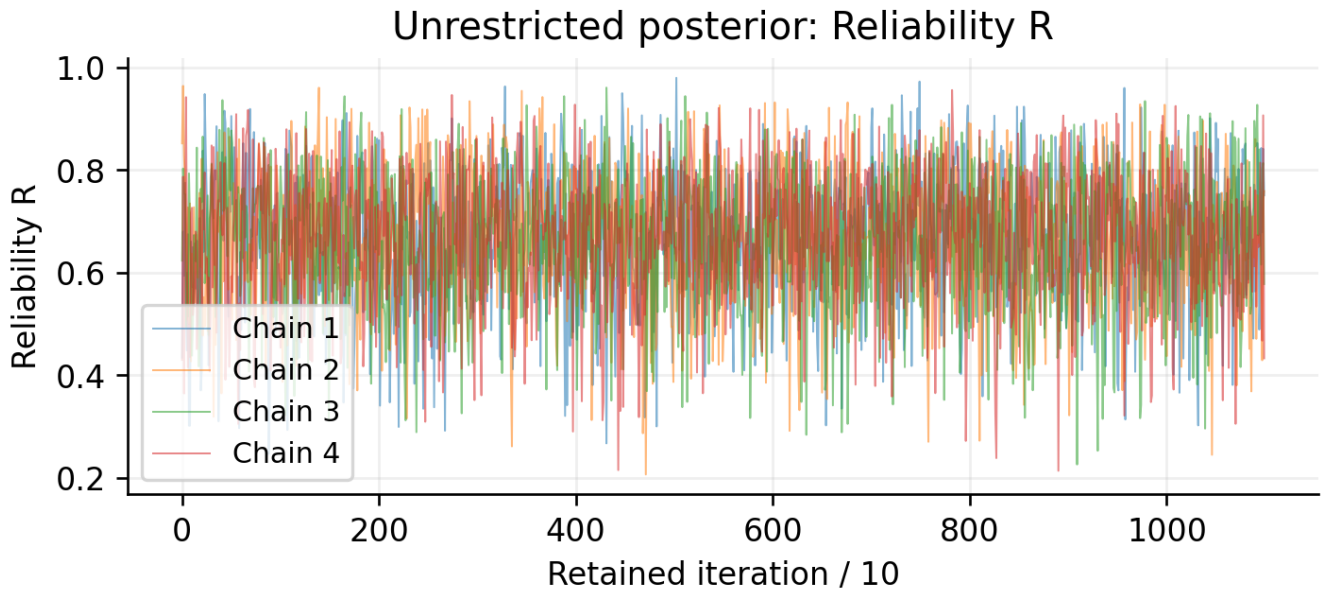


Unrestricted posterior: Stress rate b

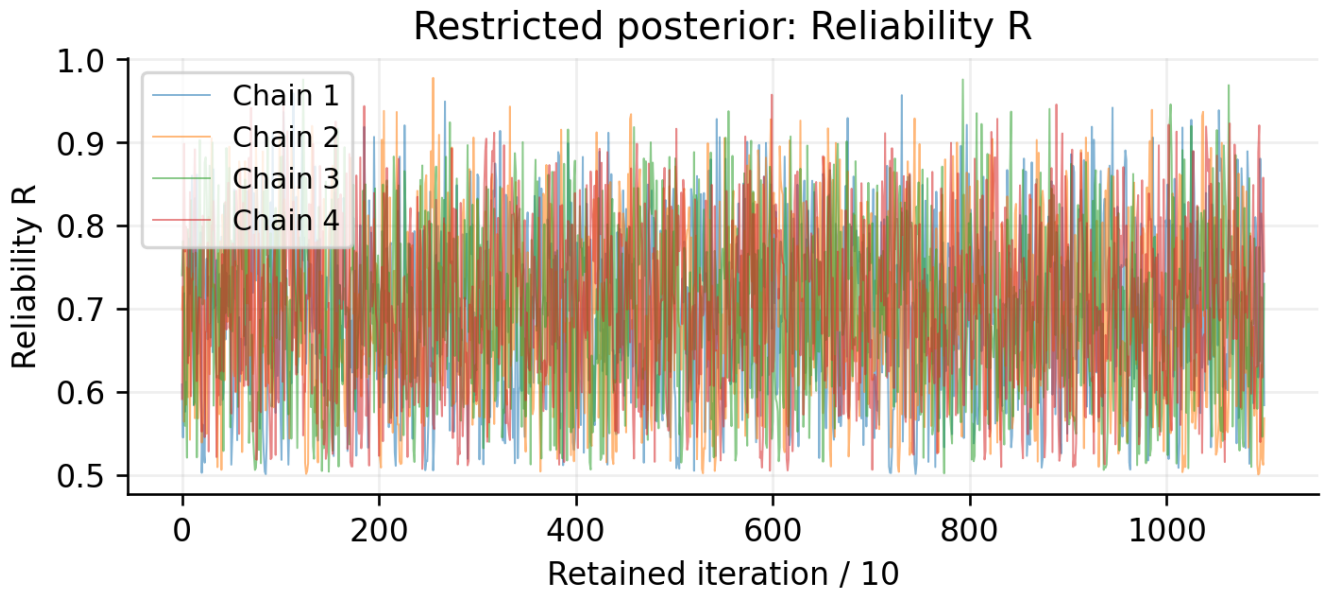


Restricted posterior: Stress rate b

Four actual retained chains for each parameter and reliability, shown every tenth iteration for readability.

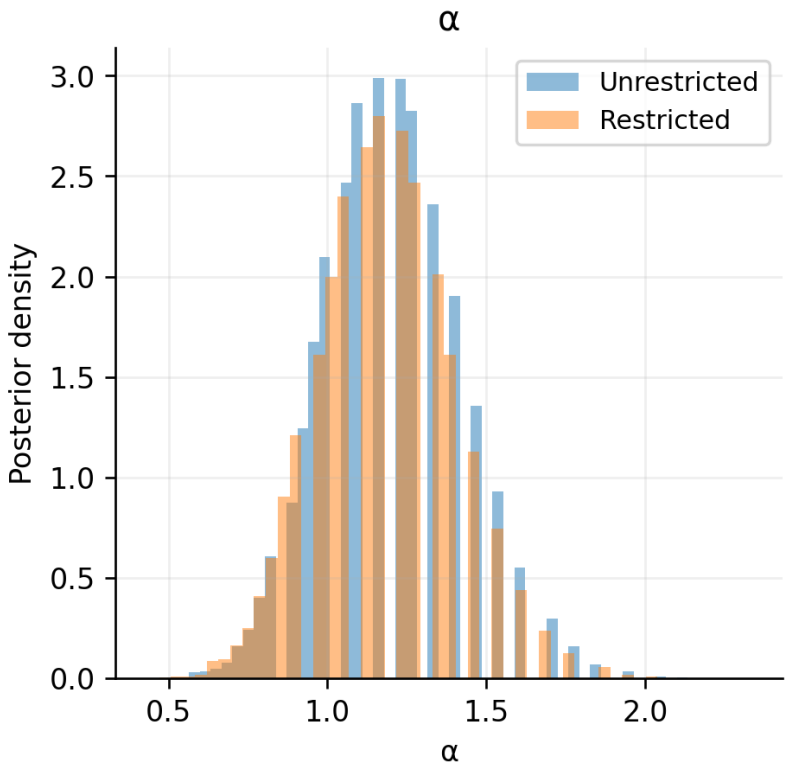


Unrestricted posterior: Reliability R

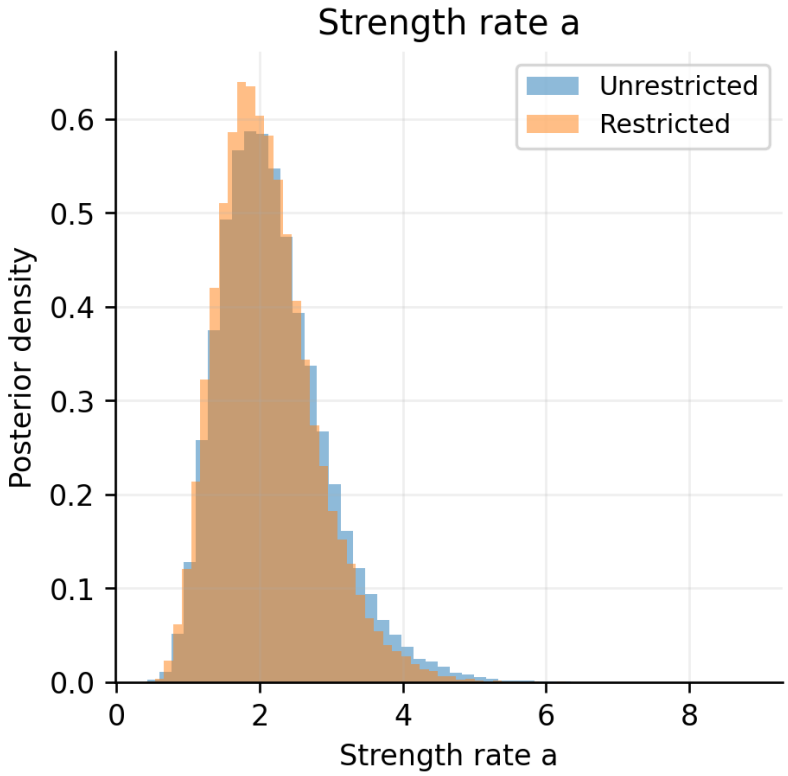


Restricted posterior: Reliability R

Marginal parameter distributions from conditional-Gamma posterior draws on the quadrature grid.

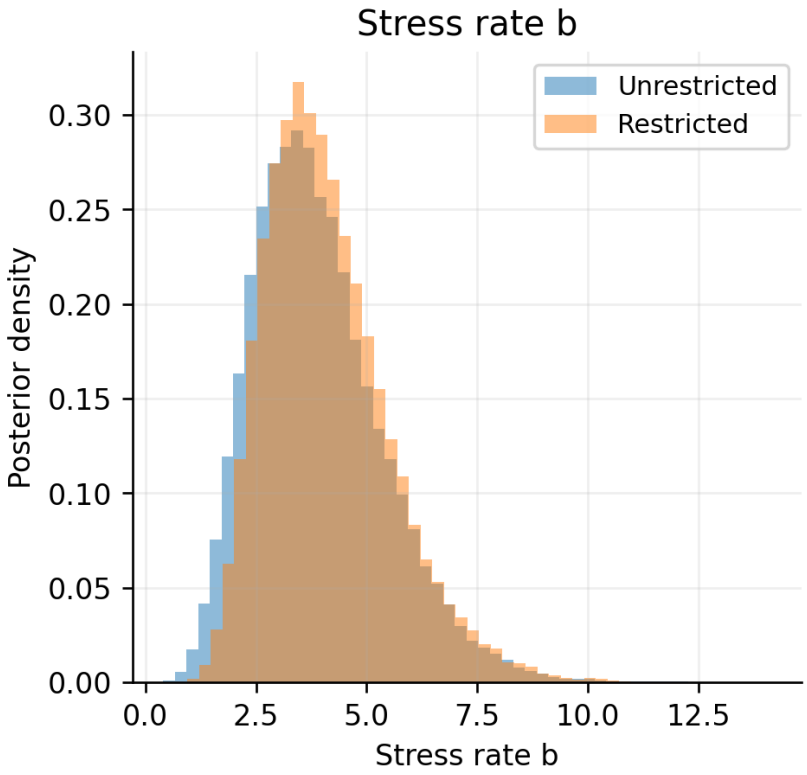


α



Strength rate a

Marginal parameter distributions from conditional-Gamma posterior draws on the quadrature grid.

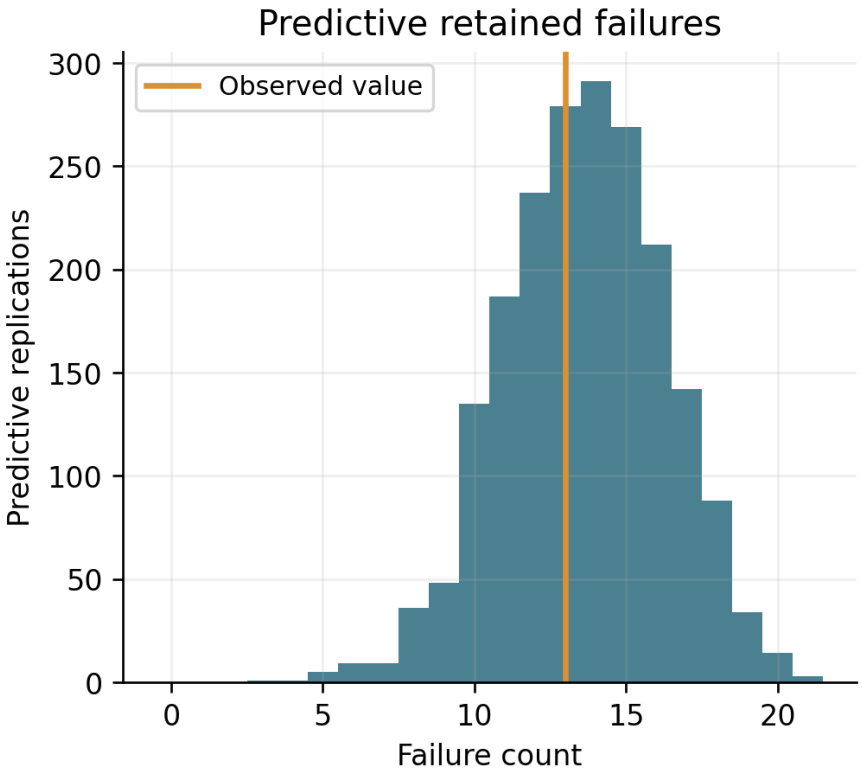


Stress rate b

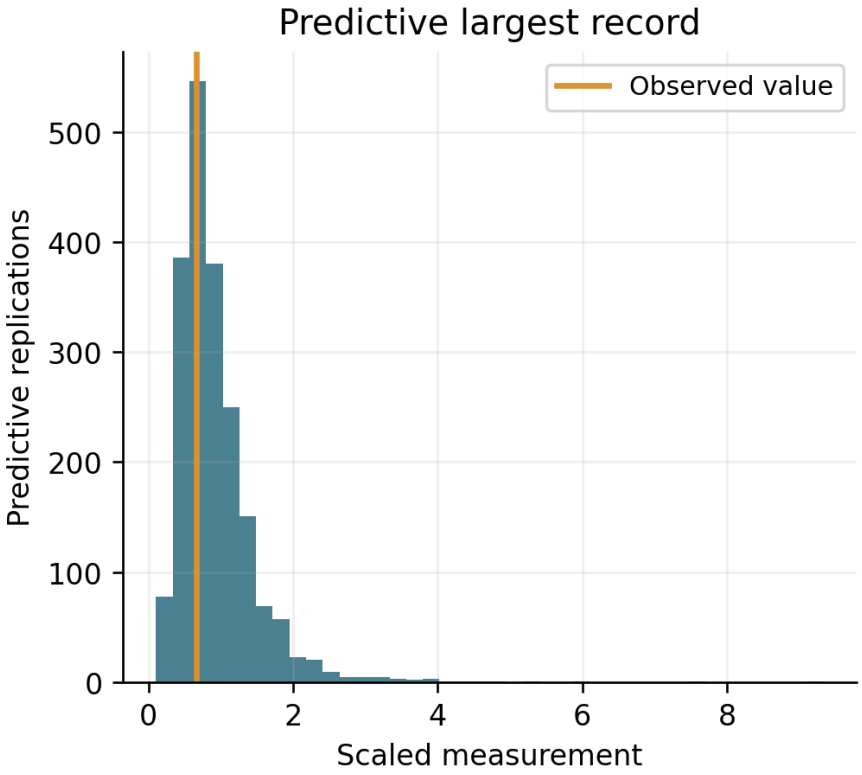
15. Sensitivity, predictive checks and interpretation

Analysis	Setting	Ordered	R	95% lower	95% upper
Prior	Primary	False	0.6656	0.3805	0.8947
Prior	Primary	True	0.6989	0.5157	0.8988
Prior	Broader total rate	False	0.6757	0.3886	0.9018
Prior	Broader total rate	True	0.7063	0.5173	0.9053
Prior	More concentrated rates	False	0.6194	0.3639	0.8474
Prior	More concentrated rates	True	0.6609	0.5103	0.8546
Hybrid scheme	Type I	False	0.6656	0.3805	0.8947
Hybrid scheme	Type I	True	0.6989	0.5157	0.8988
Hybrid scheme	Adaptive II	False	0.5450	0.2924	0.7863
Hybrid scheme	Adaptive II	True	0.6223	0.5057	0.8047
Record design	k=1, r=4	False	0.6782	0.3564	0.9183
Record design	k=2, r=4	False	0.6454	0.3321	0.8949
Record design	k=2, r=8	False	0.7116	0.4479	0.9126
Record design	k=3, r=6	False	0.5856	0.3110	0.8393

Posterior prediction under the actual observation protocol and variability across successful random acquisition orders.

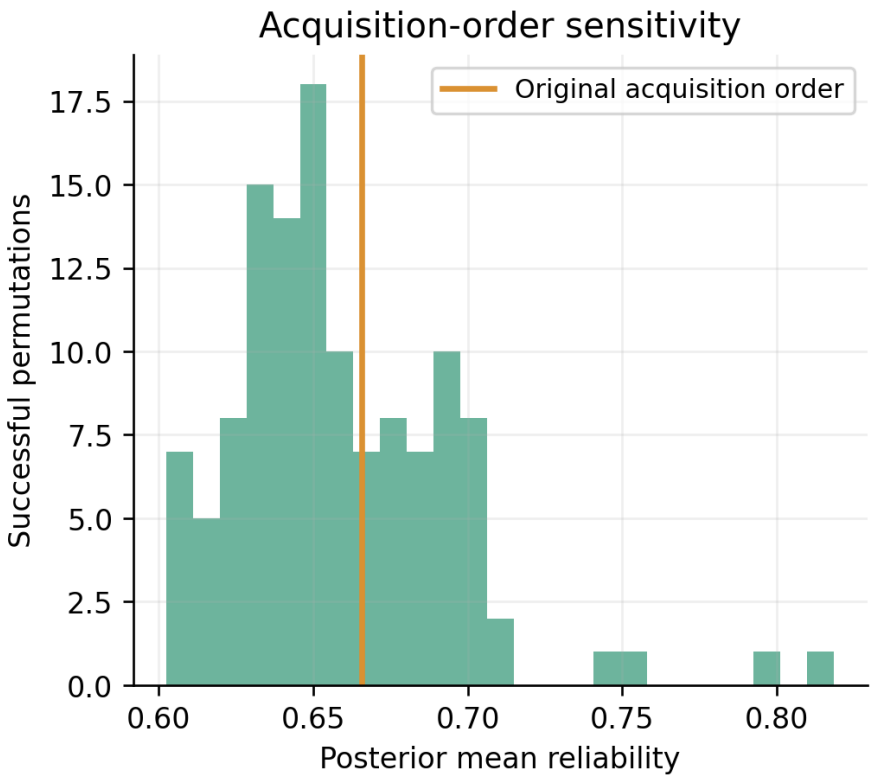


Predictive retained failures



Predictive largest record

Posterior prediction under the actual observation protocol and variability across successful random acquisition orders.



Acquisition-order sensitivity

Prior sensitivity uses fixed alternatives; hybrid sensitivity compares min-time Type-I and adaptive Type-II protocols; record sensitivity varies k and the retained count. Acquisition-order permutations that fail to reach the prescribed count are explicitly counted. The histogram conditions on successful permutations and is a sensitivity description, not an extra iid sampling experiment.

Predictive checks compare the retained failure count and largest stress record with regenerated experiments. Posterior predictive tail probabilities are descriptive checks rather than uniformly distributed significance-test p-values.

16. Literature review and source-use decisions

Newer (2025), Scientific Reports 15, 43820. DOI 10.1038/s41598-025-30570-9

Supplied 26-page full PDF. IER model with block adaptive Type-II censoring and upper k-records. Used for the combined sampling framework and performance objectives. Weibull derivations are developed separately. Its unconditional binomial expression is replaced by the appropriate conditional common-stress integral.

Sultana et al. (2026), Communications in Statistics - Theory and Methods 55(11), 3436-3459. DOI 10.1080/03610926.2025.2579720

Supplied 25-page full PDF. Common-shape Weibull block Type-I censoring, ordered rates and dependent gamma-Dirichlet priors. Used for parameterization, profiling, pooling and joint-prior construction. Table 10 supplies the jute data. Progressive withdrawals and upper records extend its design.

Newer and Alanazi (2026), Journal of Mathematics, 5561353. DOI 10.1155/jom/5561353

Supplied 17-page full PDF. Generalized exponential ranked-set/record model and stochastic system size. Used for its conditional reliability integral, posterior reliability expectations and bootstrap methodology. Section 5.1.1 supplies carbon strengths. Its lower-record ranked-set likelihood, truncated-binomial component count and empirical Bayes procedures are outside this assignment.

All five supplied articles were extracted page by page; relevant mathematical and application sections were reviewed. Original PDFs and extracted text remain in the project folder. The full source-use register is also available as literature_review.md.

17. Other supplied articles and access limitation

Hasaballah et al. (2024), AIP Advances 14, 045111. DOI 10.1063/5.0201467

Supplied 13-page PDF. Generalized inverted exponential unified hybrid inference. Used as methodological background for survival/hazard displays, Bayesian loss and interval reporting. Its distribution, unified stopping scheme and COVID-19 data are not substituted for this Weibull model.

Yeganeh et al. (2025), Engineering Applications of Artificial Intelligence 141, 109652. DOI 10.1016/j.engappai.2024.109652

Supplied 24-page PDF. Normal-ratio attention control charts and Ethereum backtesting. Examined and excluded from the assigned reliability calculations. No trading data or attention/control-chart procedure is used.

Kumari et al. (2025), Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability 239(3), 596-608. DOI 10.1177/1748006X241247866

Assigned SAGE reference. Publisher abstract accessed, but full text was not supplied and the full-PDF request was unavailable. Used only for abstract-level context on block progressive Type-II censoring and hierarchical inference. Full-text review of this reference remains access-limited; no inaccessible derivation is claimed to have been reproduced.

Xia et al. (2009), Study on the breaking strength of jute fibres using modified Weibull distribution, Composites Part A 40(1), 54-59. Jute measurements are taken from the supplied Sultana et al. reproduction, not a directly obtained original experimental file. Nichols and Padgett (2006), A Bootstrap Control Chart for Weibull Percentiles, Quality and Reliability Engineering International 22(2), 141-151, DOI 10.1002/qre.691. Carbon data are taken from the supplied Newer and Alanazi reproduction. Both transcription sources are recorded explicitly.

18. Findings, limitations and deliverable map

The complete 15 mm and 20 mm samples give calibrated Weibull K-S p-values 0.026 and 0.078. The common-shape likelihood-ratio statistic is 0.208, with bootstrap p-value 0.620. The 15 mm calibrated test rejects Weibull adequacy at the 5% level, so the jute estimates are reported with a model-fit limitation. The additional carbon-fibre halves have calibrated K-S p-values 0.585 and 0.101; their common-shape bootstrap p-value is 0.678. These are compatible with the fitted Weibull model at this sample size, without proving it true. Their unrestricted and restricted posterior reliability means are 0.3623 and 0.5653, respectively.

The retrospective Type-I experiment retains 13 strength failures and 6 upper 2-records. The unrestricted MLE reliability is 0.7526; its log-ratio 95% confidence interval is [0.4445, 0.9446]. The unrestricted posterior mean is 0.6656, with a 95% equal-tail credible interval [0.3805, 0.8947]. The restricted MLE is 0.7526; the restricted posterior mean is 0.6989, with credible interval [0.5157, 0.8988]. The restricted posterior excludes reliability below one half by construction, so this shift reflects the assumed rate order as well as the measurements.

Across 14,400 simulated datasets, all four estimators were compared against fixed truths and all raw outputs were retained. Scenario-level tables include Monte Carlo uncertainty. Under a false order restriction, the true reliability is below one half while restricted estimates cannot be; the resulting bias and low coverage are a structural consequence rather than an implementation error. At an equality boundary, a continuous truncated posterior has all interval mass above one half, so an equal-tail restricted credible interval need not include the boundary truth. This is why boundary coverage is reported explicitly.

The largest sampler split R-hat is 1.0009 and the minimum effective sample size is 3977. Sampler means are checked against quadrature with their Monte Carlo standard errors. The maximum real-data quadrature refinement difference in a mean is 1.4e-14. The model remains conditional on common shape, homogeneous block populations, noninformative withdrawals, shared stress and the laboratory interpretation of the comparison group. Acquisition-order sensitivity and prior sensitivity should be retained when presenting the estimates.

All five assigned requirements are implemented. The SAGE reference was available at abstract level only; full-text review of that unsupplied article remains access-limited. All five supplied PDFs were examined and their uses or exclusions are recorded in the literature review.

Main artifact: Vaidik_Maheshwari_Anant_Gupta_DA2141.ipynb. A readable HTML export retains outputs. reports/derivations.md provides the complete equations and assumptions; reports/literature_review.md records source use and exclusions. data contains transcriptions and observation designs; results contains actual computed tables, raw simulations and sampler chains; figures contains PNG and PDF charts. analysis.py and build_notebook.py support reproducibility. Python code contains no comments.